

MIPI DSI-2 IP User Guide

Updated for Quartus® Prime Design Suite: **25.3**

IP Version: **2.0.0**



Online Version



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1. Overview

1.1. About the MIPI DSI-2 IP

The MIPI DSI/DSI-2 specification define a video interface standard commonly used to transmit video from a host to a display screen. Its key features are low power utilization, and formatting/timing that allows display implementation using a simple receiver without frame buffering.

The MIPI DSI-2 IP transmits video from Agilex™ 3 and Agilex 5 FPGAs to display screens via MIPI DSI-2 over the MIPI D-PHY physical layer. The IP receives video in the Altera Video Streaming Protocol (full variant) over MIPI D-PHY interfaces of 1-4 lanes in all data types defined in the DSI v1.3 and DSI-2 v2.0 specifications. The output interface is compliant with the PHY-Protocol Interface (PPI) defined within the MIPI D-PHY specification and is intended to directly interface with the Altera MIPI D-PHY IP.

1.1.1. IP Release Information

IP versions are the same as the Quartus® Prime Design Suite software versions up to v19.1. From Quartus Prime Design Suite software version 19.2 or later, IP cores have a new IP versioning scheme.

The IP version (X.Y.Z) number may change from one Quartus Prime software version to another. A change in:

- X indicates a major revision of the IP. If you update your Quartus Prime software, you must regenerate the IP.
- Y indicates the IP includes new features. Regenerate your IP to include these new features.
- Z indicates the IP includes minor changes. Regenerate your IP to include these changes.

Table 1. IP Release Information

Item	Description
IP Version	2.0.0
Quartus Prime Version	25.3
Release Date	2025.09.29

1.1.2. IP Features

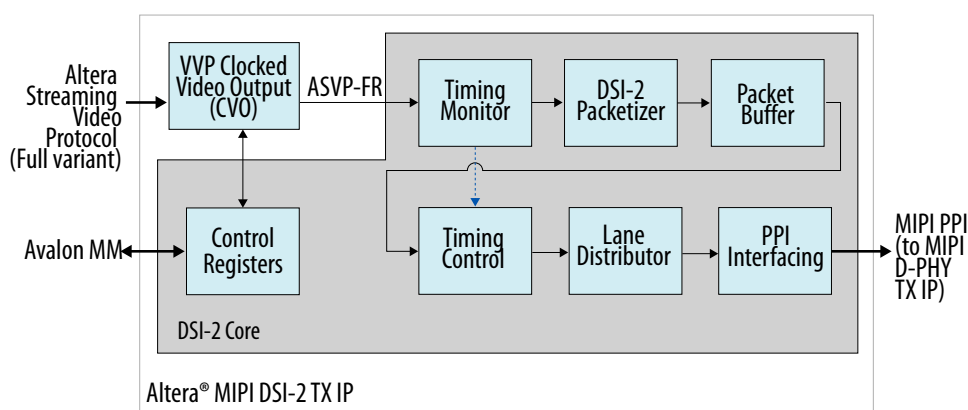
- MIPI DSI/DSI-2 transmitter protocol layer soft IP.
- 1, 2 and 4 pixels in parallel on video interface.
- 1, 2 and 4 MIPI lanes.
- Supports pixel-to-byte conversion in all video data types defined in the MIPI DSI v1.3 and MIPI DSI-2 v2.0 specifications.
- Supports video mode in unidirectional configurations up to the maximum rate supported by the FPGA.
- Supports non-burst mode with sync pulses.
- Avalon® memory-mapped interface for register access.
- AMBA AXI4-Stream interface compliant to Altera Video Streaming (Full) protocol for video data streaming.
- User-configurable video timing.
- Provides optional synchronous I/O and fixed latency modes for enhanced video timing accuracy.
- Optional ECC/CRC insertion and optional register interface.

The following functions are not supported:

- Non-burst mode with sync events and burst mode.
- Bus turn-around.
- Command mode.
- Multiple virtual channels.
- Latency Reduction Transport Efficiency (LRTE).

1.1.3. IP High-level Block Diagram

Figure 1. IP Block Diagram



1.1.4. Licensing the IP

The IP is included with the Quartus Prime Pro Edition software and is available in the IP Catalog and Platform Designer. A specific IP license for the IP is required, which is available from the Altera Self-Service Licensing Center.

After logging into the Licensing Center:

- Select **Sign up for Evaluation or No-Cost Licenses**.
- Add the MIPI DSI/CSI-2 FPGA IP license (IP-MIPI-DSI-CSI).
- Follow the prompts to generate a license for your system.

1.1.5. Device Family Support

The following table defines the IP core support levels for Altera FPGA device families and states the level of support available in the current release:

Table 2. Altera FPGA IP Core Device Support Levels

Device Support Level	Definition
Advance	The IP core is available for simulation and compilation for this device family. Timing models include initial engineering estimates of delays based on early post-layout information. The timing models are subject to change as silicon testing improves the correlation between the actual silicon and the timing models. You can use this IP core for system architecture and resource utilization studies, simulation, pinout, system latency assessments, basic timing assessments (pipeline budgeting), and I/O transfer strategy (datapath width, burst depth, I/O standards trade-offs).
Preliminary	The IP core is verified with preliminary timing models for this device family. The IP core meets all functional requirements but might still be undergoing timing analysis for the device family. It can be used in production designs with caution.
Final	The IP core is verified with final timing models for this device family. The IP core meets all functional and timing requirements for the device family and can be used in production designs.

Table 3. MIPI DSI-2 IP Core Device Family Support

This table shows the level of support offered by the MIPI DSI-2 IP for each Altera FPGA device family.

Device Family	Support
Agilex 3 (A3CW and A3CY devices) ⁽¹⁾	Preliminary
Agilex 5 E-Series	
Agilex 5 D-Series	

1.1.6. Device Speed Grade Support

The IP supports the following MIPI data rates:

- Up to 2.5Gbps per lane on all Agilex 3 and Agilex 5 E-series Group B FPGAs.
- Up to 3.5Gbps per lane on Agilex 5 E-Series Group A and Agilex 5 D-Series devices on all device speed grades.

⁽¹⁾ The MIPI DSI-2 IP does not support the Agilex 3 A3CZ devices.

The following table shows the recommended speed grade according to the frequency of the AXI4-Stream video input clock `clk`.

Table 4. Device Speed Grade Support

Series	Maximum <code>clk</code> Frequency (MHz)	Speed Grades
Agilex 5 D-Series	500	-3, -2, -1
	550	-2, -1
	600	-1
Agilex 5 E-Series (Group A)	450	-3, -2, -1
	500	-2, -1
	550	-1
Agilex 5 E-Series (Group B)	300	-6, -5, -4
	350	-5, -4
	400	-4
Agilex 3	300	-7, -6
	350	-6

1.1.7. IP Resource Utilization

The following tables show the typical device resources for selected configurations using the Quartus Prime Pro Edition software. The number of ALMs and logic registers are rounded to the nearest 50. The number stated for M20K memory blocks are not rounded. Tables are provided with utilization when runtime control via Control and Status Registers is enabled/disabled.

The below resource utilization is similar for all data types except for RGB_30bit and RGB_36bit. RGB_30bit or RGB_36bit support increases resource utilization by 550 ALMs and one M20K.

Table 5. MIPI DSI-2 Resource Utilization (without runtime control)

Configuration		ALMs	Logic Registers	M20K Memory Blocks
Data Lanes	Pixel-in-Parallel			
1-2	1	500	1050	12
1-2	2	950	1900	14
1-2	4	1550	2950	18
4	1	950	1450	20
4	2	1350	2250	22
4	4	1950	3300	26

Table 6. MIPI DSI-2 Resource Utilization (with runtime control)

Configuration		ALMs	Logic Registers	M20K Memory Blocks
Data Lanes	Pixel-in-Parallel			
1-2	1	3100	5450	12
1-2	2	3350	6300	14
1-2	4	4150	7350	18
4	1	3550	5850	20
4	2	3950	6650	22
4	4	4550	7700	26

1.2. About the Document

1.2.1. Target Audience

This document is intended for the following users:

- System Architect
- Interface Designer
- FPGA Application Designer
- Software Developer
- Test Engineer

1.2.2. Conventions Used in the Document

This document provides examples from both an operating system command-line interface (CLI) and a graphical user interface (GUI), and uses the following typographical conventions:

Table 7. Conventions

Convention	Description
<code>codeblock</code>	Indicates CLI text, such as command prompts.
boldface	Text Indicates GUI fields or dialog box title.
<code>monotype</code>	Indicates commands or file names.

1.2.3. Acronyms

This section provides a list of acronyms used in this document and their expanded forms:

Table 8. Acronym

Acronym	Expanded Form
AMBA	Advanced Microcontroller Bus Architecture
ASVP-FV	Altera Streaming Video Protocol (Full Variant)
ASVP-FR	Altera Streaming Video Protocol (Full Raster)
AXI	Advanced Extensible Interface (AMBA)
CRC	Cyclic Redundancy Checksum
CSR	Control and Status Register
CVO	Clocked Video Output (Altera VVP IP)
DSI	Display Serial Interface
DT	Data type (MIPI DSI)
ECC	Error Correction Code
HS	High-Speed Mode (MIPI D-PHY)
LP	Low-Power Mode (MIPI D-PHY)
MIPI	Mobile Industry Processor Interface
PPI	PHY-Protocol Interface (MIPI D-PHY)
RX	Receiver
TX	Transmitter
VVP	Altera Video and Vision Processing Suite

1.2.4. Reference Documents and Training

Table 9. Reference Documents

Document	Document No. / Location / URL
Altera FPGA Knowledge Base	https://www.intel.com/content/www/us/en/support/programmable/kdb-filter.html#q=dsi
MIPI D-PHY IP User Guide: Agilex 3 and Agilex 5 FPGAs	https://www.intel.com/content/www/us/en/docs/programmable/817561.html
Agilex 3 FPGA and SoC FPGA Design Hub	https://www.intel.com/content/www/us/en/support/programmable/support-resources/guided-journey/agilex3/design-hub.html
Agilex 5 FPGA and SoC FPGA Design Hub	https://www.intel.com/content/www/us/en/support/programmable/support-resources/guided-journey/agilex5/design-hub.html
MIPI DSI Specification v1.3	https://www.mipi.org/specifications/dsi
MIPI DSI-2 Specification v2.0	https://www.mipi.org/specifications/dsi-2
MIPI D-PHY Specification v2.5	https://www.mipi.org/specifications/d-phy
Altera FPGA Streaming Video Protocol Specification	https://www.intel.com/content/www/us/en/docs/programmable/683397/current/about-the-intel-fpga-streaming-video.html

1.3. Applicable Applications

The MIPI DSI-2 IP can be used in systems where a video output to a MIPI DSI/DSI-2 compatible display is required. Additionally, it can be used in video bridging applications where MIPI DSI/DSI-2 is used to provide a low-complexity video interface between the FPGA and another device within a system.

1.4. Key Concepts

The MIPI DSI and MIPI DSI-2 specifications describe a low-complexity interface that is primarily intended to send video to display modules with limited compute power and buffering. The video is presented on the interface with timing intended to ensure that the video data arrives at the ideal time for display. This reduces or removes the need for the receiver to store or process the video data.

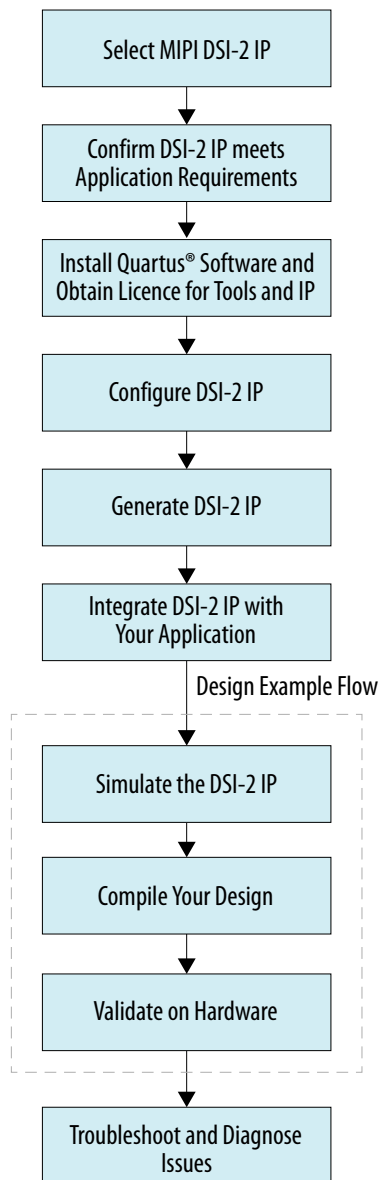
The standards are interchangeable and provide a means of packing the pixel data, along with video sync information, and sending it over the physical interface with the required video timing.

The specifications use the MIPI D-PHY physical interface to carry the packetized data. The IP is designed to interface directly with the Altera MIPI D-PHY IP, using the standard PHY-Protocol Interface (PPI) defined in the D-PHY specification.

1.5. IP Design Flow

The following flowchart illustrates the MIPI DSI-2 IP design flow. The IP can generate a simulation testbench and MIPI subsystem for synthesis from the IP parameter editor. For more information, refer to Design Example Tab.

Figure 2. IP Design Flow Diagram



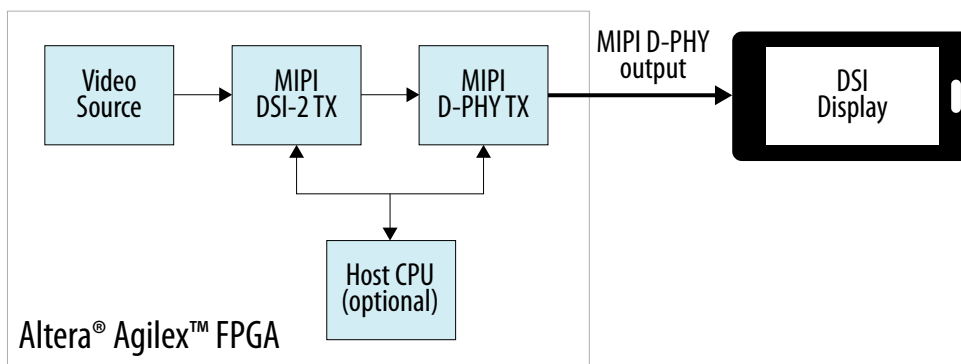
Related Information

[Design Example Tab Options](#) on page 27

1.5.1. System Level Block Diagram

A typical system uses the MIPI DSI-2 TX IP in conjunction with the MIPI D-PHY IP to present a MIPI D-PHY interface to an external display.

Figure 3. System Block Diagram



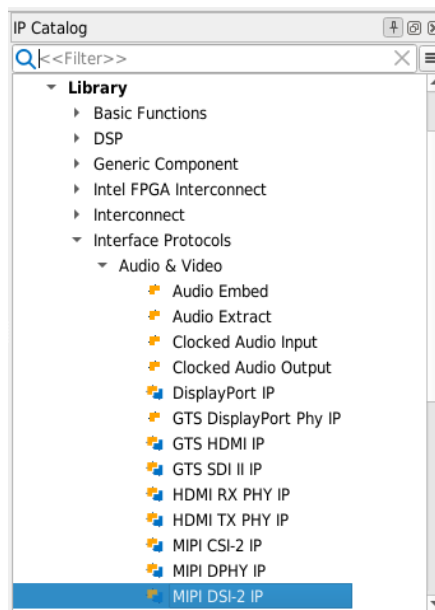
2. Quick Start Guide

2.1. Synthesizing the Design Example

Follow these steps to create a MIPI DSI-2 TX design for synthesis:

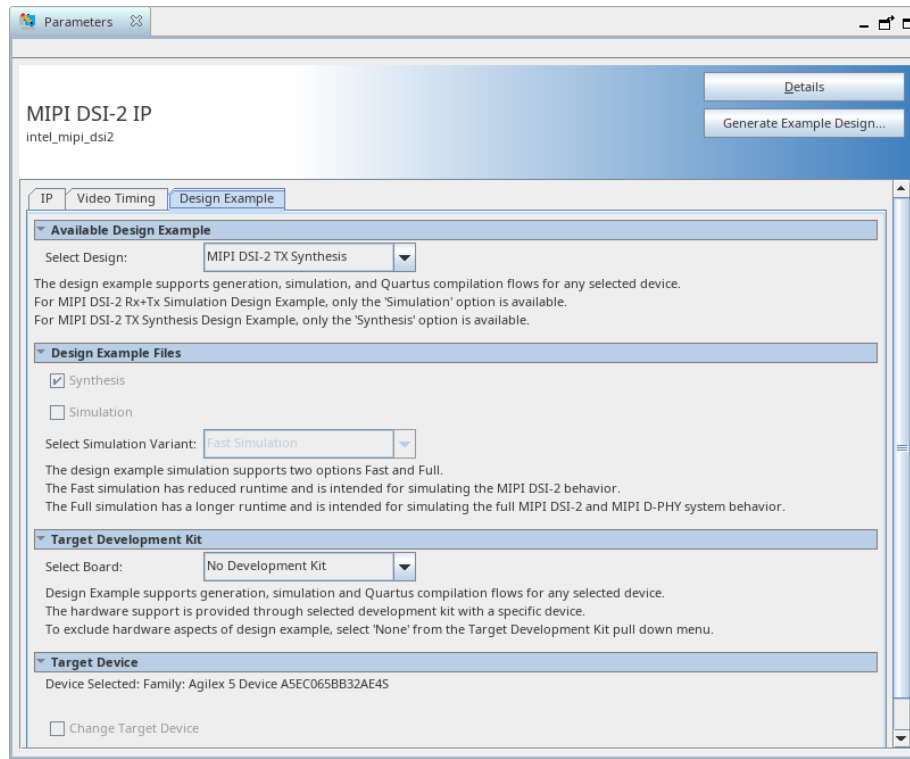
1. Create a new project in Quartus Prime Pro Edition software and specify a device for your design (Refer to [Device Family Support](#) for the supported devices).
2. In the **IP Catalog**, select **Library** ► **Interface Protocols** ► **Audio & Video** ► **MIPI DSI-2 IP**.

Figure 4. MIPI DSI-2 IP in the IP Catalog



3. Set up the parameters for the MIPI DSI-2 IP as required for your design:
 - a. Configure the parameters on the **Video Timing** tab according to your design. Refer *Video Timing Tab* for more information.
 - b. In the **Design Example** tab, select the **MIPI DSI-2 TX Synthesis**.

Figure 5. MIPI DSI-2 Design Example Tab



4. Click the **Generate Design Example** to generate the design.
5. Open the generated example design project in Quartus Prime Pro Edition.
6. Configure the I/O as appropriate for your development board. You may need to open the Platform Designer system and ensure that the MIPI D-PHY IP configuration matches your hardware – particularly the byte location for the enabled MIPI link.
7. Add a video source and connect to the AXI4-Stream video input to the MIPI DSI-2 IP.
8. Compile the project and test.

Note: No host processor is required to initialize or control the MIPI interface. Without a controller, the interface operates from power-up in the pre-configured mode.

The reset on the MIPI DSI-2 IP should always be asserted for at least 256 cycles of the `clk` clock input to the MIPI DSI-2 before starting operation.

Related Information

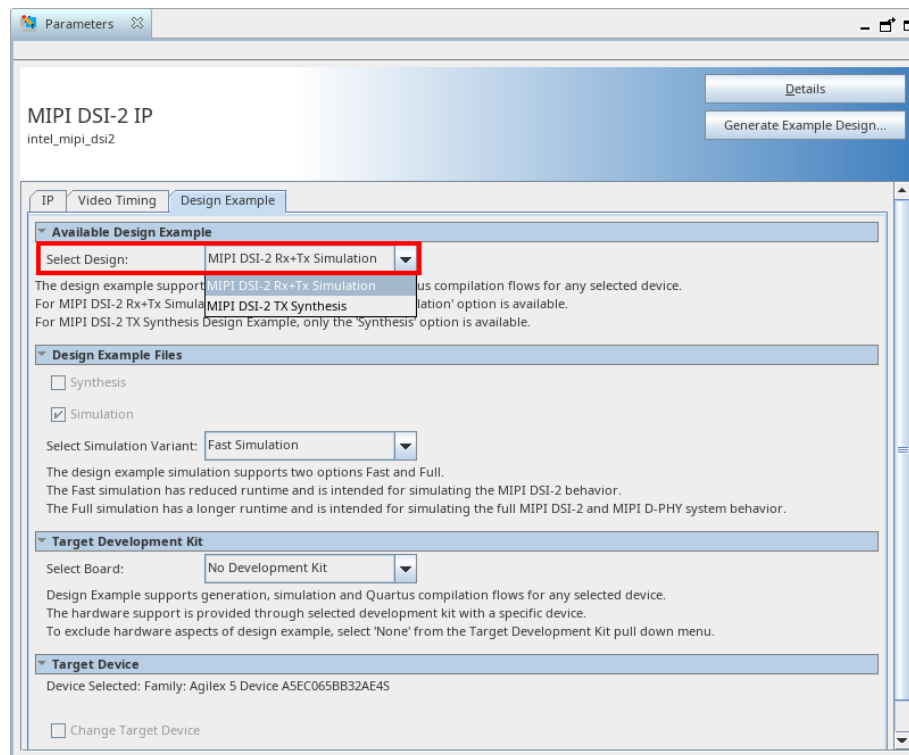
[Video Timing Tab Options](#) on page 20

2.2. Simulating the Design Example

Follow these steps to create a MIPI DSI-2 TX design for simulation:

1. Create a new project in Quartus Prime Pro Edition software and specify a device for your design (Refer to [Device Family Support](#) for the supported devices).
2. In the **IP Catalog**, select **Library** ➤ **Interface Protocols** ➤ **Audio & Video** ➤ **MIPI DSI-2 IP**.
3. Set up the parameters for the MIPI DSI-2 IP as required for your design:
 - a. Configure the parameters on the **Video Timing** tab according to your design. Refer *Video Timing Tab* for more information.
 - b. In the **Design Example** tab, select the **MIPI DSI-2 Rx + Tx Simulation**.

Figure 6. MIPI DSI-2 Design Example Tab with Simulation Option



4. Click the **Generate Design Example** to generate the design.
- Refer to the [Simulating the Design Example](#) for more information.



3. Configuring and Generating the IP

This section describes the steps required to configure and generate the IP and the design examples.

3.1. Configuring the Quartus Prime Pro Edition Project

Follow these steps to create a Quartus Prime project:

1. In the Quartus Prime Pro Edition software, click **File > New Project Wizard** to create a new Quartus Prime project. The wizard prompts you to specify a device. To open existing project, go to **step 4**.
2. Specify the device family and select the device for your design. Refer to [Device Family Support](#) for the supported devices.
3. Follow the on-screen instruction to complete the New Project creation.
4. For existing project, go to **File > Open** to open an existing Quartus Prime project.

A valid license is required to use the IP which can be obtained from the Altera FPGA Self-Service Licensing Center. Refer to [Licensing the IP](#).

3.2. Configuring the MIPI DSI-2 IP

Perform the following steps to configure the IP:

1. Select **Tools > IP Catalog** to open the IP Catalog.
2. Select **Library > Interface Protocols > Audio & Video > MIPI DSI-2 IP**.
3. Click **Add**.
4. Specify a top-level name *<your_ip>* and the folder for your custom IP variation. The IP parameter editor saves the IP variation settings in a file named *<your_ip>.ip*.
5. Click **Create**. The IP parameter editor appears.

3.2.1. Configuring the MIPI DSI-2 IP Parameters

3.2.1.1. IP Tab Options

On the **IP** tab, specify the parameters to configure the MIPI and video interface for your IP.

Figure 7. IP Tab Parameters

The screenshot shows the 'Parameters' window for the 'MIPI DSI-2 IP' (intel_mipi_dsi2). The 'IP' tab is selected, showing various configuration options:

- Video Streaming Interface Options:**
 - Pixels in parallel: 2
 - Pixels per video line: 128
 - AXI4-S TDATA width: 48 bits
 - ☒ Use common clock for video input and MIPI output
- Video Packing/Unpacking Options:**
 - MIPI DSI-2 Data Type: RGB_24bit
- MIPI D-PHY Interface Options:**
 - Number of lanes: 2
 - PPI bus width per lane: 2 bytes
- General Options:**
 - Direction: TX
 - ☒ Control and Status Registers (CSR)
 - ☒ CRC insertion
 - ☒ ECC insertion
 - ☒ Fixed latency mode
 - Buffer depth: 4096

Buttons for 'Details' and 'Generate Example Design...' are visible in the top right.

Table 10. IP Tab Parameter Description

Parameter	Range Selection	Default Setting	Description
Pixels in parallel	1, 2, 4	2	Select the number of pixels presented on each beat of the input video interface.
Pixels per video line	1–8192	128	Indicate the number of pixels on a video line. This must be set to the exact line length to be transmitted.
Use common clock for video input and MIPI output	On/Off	On	When the common clock is enabled, the video input must be driven from the MIPI link clock for this interface. This allows the MIPI DSI-2 IP to operate with a single clock domain and enables the fixed latency mode option.
<i>continued...</i>			

Parameter	Range Selection	Default Setting	Description
AXI4-S TDATA width	1–96	N/A	This parameter indicates the calculated width of the video TDATA input based on the IP configuration. This value is fixed and cannot be changed.
MIPI DSI-2 data type	• RGB 36bit • RGB_30bit • RGB 24bit • RGB 18bit (Packed) • RGB 18bit (Loose) • RGB 16bit • YCbCr 24bit • YCbCr 20bit (Packed) • YCbCr 20bit (Loose) • YCbCr 16bit • YCbCr 12bit	RGB 24bit	Select the MIPI DSI-2 data type to be used when video data is packed into MIPI DSI-2 packets. The pixel data received on the AXI4-Stream interface must correspond to this data type.
Number of lanes	1, 2, 4	2	MIPI lanes on the external MIPI D-PHY interface. This must correspond to the MIPI D-PHY IP configuration.
PPI bus width	2	2	This indicates the number of bytes per lane on the data interface on the PPI. This value is fixed and cannot be changed.
Direction	Tx	Tx	This option is fixed to TX. The IP is currently only available as a transmitter.
Control and Status Registers (CSRs)	On/Off	On	When Control and Status Registers are off, the control and status register interface is not included. Disabling this option reduces the IP resource utilization but the CVO and Video Timing Generator cannot be dynamically reconfigured.
CRC insertion	On/Off	On	Cyclic redundancy check (CRC) processing. When CRC insertion is disabled, the IP does not calculate long packet payload CRC and it sets the fixed value zero in the checksum field. Disabling this parameter reduces the resource utilization in your system architecture where it is not required. <i>Note:</i> If you choose to disable CRC insertion for your system, the output from the IP will not comply with the MIPI DSI-2 specification.
ECC insertion	On/Off	On	Error correction code (ECC) processing. When ECC insertion is disabled, the IP does not calculate the header ECC code and sets the fixed value zero in the ECC field in all packet headers. Disabling this parameter reduces the resource utilization in your system architecture where it is not required. <i>Note:</i> If you choose to disable ECC insertion for your system, the output from the IP will not comply with the MIPI DSI-2 specification.
Fixed latency mode	On/Off	On	The fixed latency mode adheres strictly to the video timing configured in the IP. When disabled, the video is passed with minimum latency. So, the MIPI output follows the line and frame intervals configured. The sync locations may vary slightly within lines, depending on the timing configured. Enabling this option slightly increases logic and memory utilisation. <i>Note:</i> Only use this option when common video input and MIPI clocks are enabled.
Buffer depth	128– 65536	4096	Configure the internal data buffer depth. This must be enough to hold at least 1.2 video lines. If the incoming video data is very bursty, increasing this option enhances resilience to data underflow errors within the IP. To assist you in selecting the

Parameter	Range Selection	Default Setting	Description
			appropriate buffer size, the currently selected buffer depth is translated into video lines and displayed by the IP. It also can be calculated using this formula: $buffer_depth_{lines} = \frac{buffer_depth_{words} \times PPI_bus_width \times MIPI_lanes}{packed_bits_per_pixel \times pixels_per_line}$

3.2.1.2. Video Timing Tab Options

On the **Video Timing** tab, specify the parameters to set the timing of the video on the MIPI D-PHY interface.

Figure 8. Video Timing Tab Parameters

The screenshot shows the 'Parameters' window for the 'MIPI DSI-2 IP' (intel_mipi_dsi2). The 'Video Timing' tab is selected. The window includes a 'Details' button and a 'Generate Example Design...' button. The configuration is divided into three sections: Dimensions, Blanking Timing, and Sync Timing.

Section	Parameter	Value
Dimensions	Default value for HTOTAL:	2176
	Default value for VTOTAL:	100
Blanking Timing	Default value for HB_END:	2048
	Default value for V1B_START:	0
	Default value for V1B_END:	4
Sync Timing	Default Value for HS_START:	52
	Default Value for HS_END:	100
	Default Value for V1S_VSTART:	2
	Default Value for V1S_HSTART:	52
	Default Value for V1S_VEND:	3
	Default Value for V1S_HEND:	52

Table 11. Video Timing Tab Parameter Description

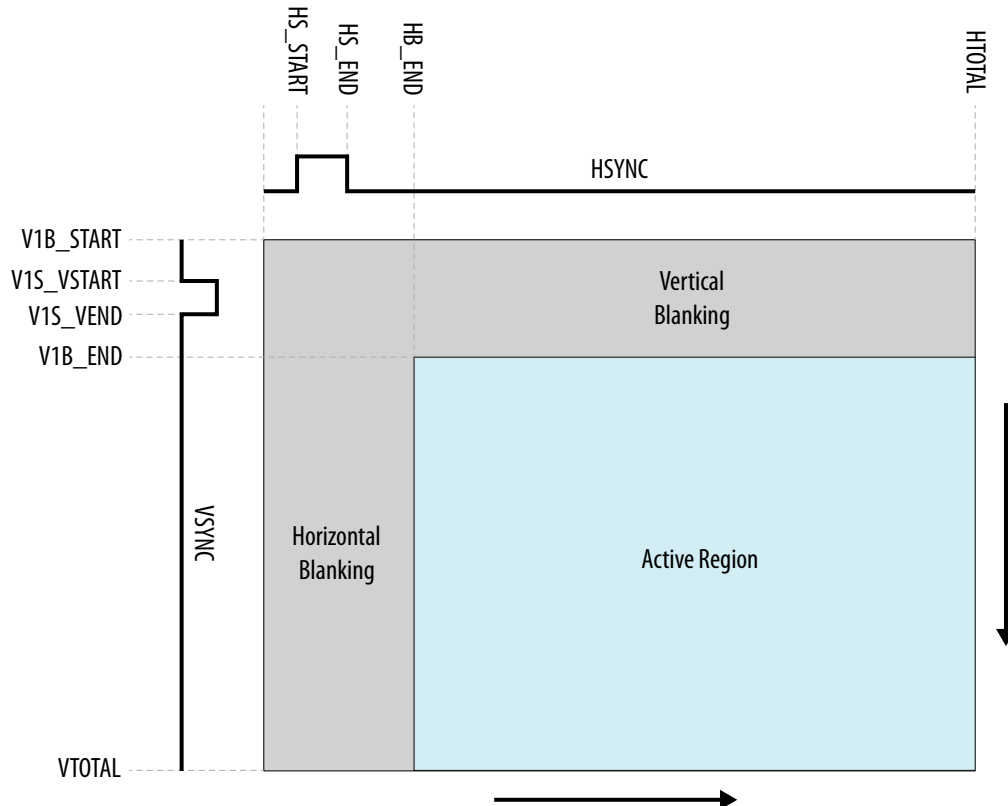
Parameter	Values	Description
Video timing configuration values	Manual	This option selects the video timing configuration in operation. The IP currently only supports manual mode where all timing parameters are manually configured.
Validate video timing parameters	On/Off	When enabled, the video timing parameters are checked to ensure that they meet the configuration requirements (Refer to Configuration Requirements). If this is disabled, the checks are not performed and the IP is not guaranteed to function correctly.
Default value for HTOTAL	0–65536	Total horizontal line length in pixels, including active and blanking regions.
Default value for VTOTAL	0–65536	Total vertical picture size in video lines, including active and blanking regions.
Default value for HB_END	0–65536	Horizontal blanking period in pixels. The difference between HTOTAL and this value must equal the horizontal active line length in pixels. When the YCbCr 12bit data type is used, the difference must be equal to half of the horizontal active line length.
Default value for V1B_START	0–65536	Line number of first line of vertical blanking. <i>Important:</i> This value must be set to zero.
Default value for V1B_END	0–65536	Line number of first line of active region. The interval between V1B_END and VTOTAL must be equal to the number of active lines per video frame.
Default value for HS_START	0–65536	Horizontal position of the start of horizontal sync pulse. This must be within the horizontal blanking region, HB_END.
Default value for HS_END	0–65536	Horizontal position of the end of horizontal sync pulse. This value must be between HS_START and HB_END.
Default value for V1S_VSTART	0–65536	Vertical position of the start of the vertical sync pulse. This value must be after V1B_START and before V1B_END.
Default value for V1S_HSTART	0–65536	Horizontal position of the start of the vertical sync pulse. This value must be within the horizontal blanking region (less than HB_END).
Default value for V1S_VEND	0–65536	Vertical position of the end of the vertical sync pulse. This value must be after V1S_VSTART and before V1B_END.
Default value for V1S_HEND	0–65536	Horizontal position of the end of the vertical sync pulse. This value must be between V1S_HSTART and HB_END.

3.2.1.2.1. Configuration Requirements

Configuration of the video timing requires a detailed understanding of the requirements of the intended display in your system and Altera recommends that you consult your display manufacturer for guidance. When entering the horizontal timing parameters, it is important to calculate the values based on the video input clock rate. This clock rate may be different from the datasheet values.

You must consider the following factors described in this section when determining the timing configuration to ensure that the system is viable. Otherwise, the output video may be corrupted or lose synchronization. In some cases, it may be necessary to increase the MIPI data rate or increase the lane count on the MIPI D-PHY interface to ensure that sufficient bandwidth is available to carry the required video stream.

Figure 9. Video Timing Parameters



The timing parameters are defined in the same way as the Video Timing Generator IP in the Altera Video and Vision Processing Suite. Note that all horizontal values are defined in horizontal units rather than clock cycles. In the YCbCr 12-bit format, the horizontal unit is a pair of pixels, due to the way that 4:2:0 data is carried. In all other formats, the horizontal unit is a single pixel.

For example, if the IP is configured with 2 pixels-in-parallel on the AXI4-Stream video interface with the data type RGB 24-bit and HTOTAL=2200, then the line interval is 1100 clock cycles. This should be considered when calculating line intervals and frame rates.

Important: The configuration of all horizontal timing parameters (HTOTAL, HB_END, HS_START, HS_END, V1S_HSTART, V1S_HEND) must be an integer value that is the multiple of the configured number of pixels in parallel at the input. For example, if the IP is configured with 2 pixels in parallel, then both HB_END and HTOTAL must be a multiple of 2.

Also, the relationship between HB_END and HTOTAL must always be:

$$HB_END = HTOTAL - active_line_length$$

For YCbCr 12-bit format, the relationship is:

$$HB_END = HTOTAL - \frac{active_line_length}{2}$$

Altera recommends that you calculate and observe the minimum value of HTOTAL that is supported for a given system. The following formula indicates the shortest line interval that is supported by the selected configuration of the IP.

$$HTOTAL_{min} = \left(700 + \frac{packed_pixel_width \times active_line_length}{lanes \times ppi_width} \right) \times pixels_in_parallel \times 1.1$$

If the MIPI D-PHY IP link is configured longer than the default timing interval parameters, this value requires a corresponding increase. This value is calculated and displayed in the Parameter Editor. If your HTOTAL value does not comply with this value, an error is displayed. Disabling the video timing parameter validation option avoids this error but may result in video corruption and loss of synchronization.

- Note:**
- *packed_pixel_width* is the number of bits required to insert a single pixel into the packed output stream. For example, 16 for the RGB-16bit data type. When loosely-packed data types are selected, the packed pixel width should account for the additional bits consumed in these formats. For example, YCbCr 20bit (Loose) requires 24-bits per pixel when packing the video stream with unused bits, therefore the pixel width used in these calculations must use the higher value.
 - *active_line_length* is the number of pixels per video line.
 - *lanes* is the number of MIPI lanes configured in the MIPI DSI-2 IP and MIPI D-PHY IP.
 - *ppi_width* is the bit width of the PPI interface between the MIPI DSI-2 IP and MIPI D-PHY IP, which is fixed to 16.
 - *pixels_in_parallel* is the configured pixels in parallel at the streaming video input to the MIPI DSI-2 IP.

The line interval and frame rate can be calculated from the configured timing values. The video timing generation operates on the same clock used on the AXI4-Stream video input interface. When the option for the common video input and MIPI link clocks is enabled, this clock is the MIPI D-PHY link word clock, and this clock frequency is calculated and displayed in the MIPI D-PHY IP parameter editor for the configured IP. When the option for the common clocks is disabled, this is the frequency of your supplied input clock.

$$line_interval_secs = \frac{HTOTAL}{clock_freq_{AXI-Stream} \times pixels_in_parallel}$$

$$frame_rate_Hz = \frac{clock_freq_{AXI-Stream} \times pixels_in_parallel}{HTOTAL \times VTOTAL}$$

Refactoring this formula, if your frame rate and total line count is fixed, you can calculate the required horizontal timing parameters based on these values using the following formula. Note, you must ensure that HTOTAL complies with HTOTAL_{min}:

$$HTOTAL = \frac{clock_freq_{AXI-Stream} \times pixels_in_parallel}{frame_rate_Hz \times VTOTAL}$$

It is important to consider the available bandwidth on both the AXI4-Stream and MIPI interfaces to ensure that your chosen video timing is feasible. The MIPI interface must have sufficient bandwidth to pass a line of video data within the line interval shown by the calculation above. An approximation of the configured peak data transfer rates on input and output can be calculated as follows:

$$data_rate_input_bps = packed_pixel_width \times active_line_length \times VTOTAL \times frame_rate$$

$$data_rate_output_bps = mipi_lanes \times lane_bitrate$$

For example, a system carrying YCbCr 20bit (Packed) data with a line length of 1280 pixels, 815 total lines per frame (active + blanking) at 60 frames per second has a peak input data rate of 1.25Gbps + MIPI signaling overheads.

When comparing input and output rates, the MIPI DSI-2 in-band sync signaling and MIPI D-PHY LP/HS transition duration must be taken into consideration. The above calculation for data_rate_output_bps ignores these values. The HTOTAL_{min} calculation provides a more accurate horizontal timing estimate.

3.2.1.2.2. Timing Calculation Examples

The following are examples of how the video parameters are calculated:

Example 1

Table 12. Settings Examples for Timing Parameters

Parameter	Settings
MIPI data rate	2500 Mbps
MIPI lanes	1
DSI-2 data types	YCbCr-20bit Loose (24 bits per pixel)
Video interface pixels in parallel	2
Active video dimensions	1280 pixels × 800 lines
Frame rate	90 Hz
Vertical front porch (VFP)	5 lines (e.g. panel recommendation)
Vertical back porch (VBP)	5 lines (e.g. panel recommendation)
continued...	

Parameter	Settings
Vertical sync active (VSA)	5 lines (e.g. panel recommendation)
Common video + MIPI clocks	Disabled
Video clock rate	250 MHz

1. Calculate the vertical parameters. If VFP, VSA, and VBP values are not specified by the panel manufacturer, these values can typically be a small value between 2 and 30.

Parameter	Calculation	Value
V1B_START	Fixed	0
V1S_VSTART	VFP	5
V1S_VEND	VFP + VSA	= 5+5 = 10
V1B_END	VFP + VSA + VBP	= 5+5+5 = 15
VTOTAL	V1B_END + VACTIVE	= 15+800 = 815

2. Perform sanity checks to ensure the link is able to support the required bandwidth.

$$\begin{aligned} data_rate_input_bps &= packed_pixel_width \times active_line_length \times VTOTAL \times frame_rate \\ &= 24 \times 1280 \times 815 \times 90 = 2253Mbps \end{aligned}$$

$$data_rate_output_bps = mipi_lanes \times lane_bitrate = 2500 \times 10^6 \times 1 = 2500Mbps$$

The MIPI output channel has around 11% additional bandwidth, which is likely to be sufficient to support the MIPI signaling overheads.

3. Calculate the horizontal total line length.

$$HTOTAL = \frac{clock_freq_{AXI-Stream} \times pixels_in_parallel}{frame_rate_Hz \times VTOTAL} = \frac{250 \times 10^6 \times 2}{90 \times 815} = 6816.6$$

4. Check that the calculated HTOTAL complies with HTOTAL_{min}.

$$clock_freq_{MIPi} = \frac{MIPi_data_rate}{PPI_width/2} = \frac{2500 \times 10^6}{8} = 312.5MHz$$

$$\begin{aligned} HTOTAL_{min} &= \left(700 + \frac{packed_pixel_width \times active_line_length}{lanes \times ppi_width} \right) \times pixels_in_parallel \times 1.1 \\ &\times \frac{clock_freq_{AXI-Stream}}{clock_freq_{MIPi_core_clk}} = \left(700 + \frac{24 \times 1280}{1 \times 16} \right) \times 2 \times 1.1 \times \frac{250}{312.5} = 4611 \end{aligned}$$

As the calculated HTOTAL, 6816.6, is greater than the minimum, 4611, this complies with the minimum. However, all the horizontal parameters must be an integer multiple of the video input pixels in parallel, in this case 2. Therefore, the number must be rounded to the nearest multiple of pixels in parallel, i.e. 6816.

5. Calculate the other horizontal parameters. Horizontal sync positions are often flexible and can be spread evenly within the blanking period between pixels 0 and HB_END. If these are specified by the panel manufacturer, take this information into consideration. If the video input is configured with multiple pixels in parallel, the horizontal sync positions (HS_START, HS_END, V1S_HSTART, V1S_HEND) should be multiplied by the number of pixels in parallel.

Parameter	Calculation	Value
HB_END	HTOTAL-HACTIVE	= 6816-1280 = 5536
HS_START	0 < HS_START < HB_END	e.g. 200
HS_END	HS_START < HS_END < HB_END	e.g. 400
V1S_HSTART	Align with HS_START	200
V1S_HEND	Align with HS_START	200
HTOTAL	As calculated above	6816

Example 2

Table 13. Settings Examples for Timing Parameters

Parameter	Settings
MIPI data rate	900 Mbps
MIPI lanes	4
DSI-2 data types	RGB888 (24 bits per pixel)
Video interface pixels in parallel	2
Active video dimensions	1920 pixels × 1080 lines
Frame rate	60 Hz
Vertical front porch (VFP)	10 lines (e.g. panel recommendation)
Vertical back porch (VBP)	10 lines (e.g. panel recommendation)
Vertical sync active (VSA)	5 lines (e.g. panel recommendation)
Common video + MIPI clocks	Enabled

1. Calculate the vertical parameters. If VFP, VSA, and VBP values are not specified by the panel manufacturer, these values can typically be a small value between 2 and 30.

Parameter	Calculation	Value
V1B_START	Fixed	0
V1S_VSTART	VFP	10
V1S_VEND	VFP + VSA	= 10+5 = 15
V1B_END	VFP + VSA + VBP	= 10+5+10 = 25
VTOTAL	V1B_END + VACTIVE	= 25+1080 = 1105

2. Perform sanity checks to ensure the link is able to support the required bandwidth.

$$\begin{aligned} data_rate_input_bps &= packed_pixel_width \times active_line_length \times VTOTAL \times frame_rate \\ &= 24 \times 1920 \times 1105 \times 60 = 3055Mbps \end{aligned}$$

$$data_rate_output_bps = mipi_lanes \times lane_bitrate = 900 \times 10^6 \times 4 = 3600Mbps$$

The MIPI output channel has around 18% additional bandwidth, which is likely to be sufficient to support the MIPI signaling overheads.

3. Calculate the horizontal total line length. As the common video and MIPI clock are enabled, the video AXI4-Stream clock is the same rate as the MIPI PPI interface:

$$clock_freq_{AXI-Stream} = clock_freq_{MIPI_core_clk} = \frac{MIPI_data_rate}{PPI_width/2} = \frac{900 \times 10^6}{8} = 112.5MHz$$

$$HTOTAL = \frac{clock_freq_{AXI-Stream} \times pixels_in_parallel}{frame_rate_Hz \times VTOTAL} = \frac{112.5 \times 10^6 \times 2}{60 \times 1105} = 3394$$

4. Check that the calculated HTOTAL complies with HTOTAL_{min}.

$$HTOTAL_{min} = \left(700 + \frac{packed_pixel_width \times active_line_length}{lanes \times ppi_width} \right) \times pixels_in_parallel \times 1.1 \times \frac{clock_freq_{AXI-Stream}}{clock_freq_{MIPI_core_clk}}$$

$$= \left(700 + \frac{24 \times 1920}{4 \times 16} \right) \times 2 \times 1.1 \times \frac{112.5}{112.5} = 3124$$

As the calculated HTOTAL 3394 is greater than the minimum 3124, this is a valid configuration.

5. Calculate the other horizontal parameters. Horizontal sync positions are often flexible and can be spread evenly within the blanking period between pixels 0 and HB_END. If these are specified by the panel manufacturer, take this information into consideration. If the video input is configured with multiple pixels in parallel, the horizontal sync positions (HS_START, HS_END, V1S_HSTART, V1S_HEND) should be multiplied by the number of pixels in parallel to convert from clock cycles to pixels.

Parameter	Calculation	Value
HB_END	HTOTAL-HACTIVE	= 3124-1920 = 1204
HS_START	0 < HS_START < HB_END	e.g. 200
HS_END	HS_START < HS_END < HB_END	e.g. 400
V1S_HSTART	Align with HS_START	200
V1S_HEND	Align with HS_START	200
HTOTAL	As calculated above	3394

3.2.1.3. Design Example Tab Options

On the **Design Example** tab, specify the parameters to configure and generate the design examples for the MIPI DSI-2 IP.

Figure 10. Design Example Tab Parameters

The screenshot shows the 'Parameters' window for the 'MIPI DSI-2 IP' (intel_mipi_dsi2). The 'Design Example' tab is selected. It contains the following sections:

- Available Design Example:** A dropdown menu for 'Select Design:' is set to 'MIPI DSI-2 TX Synthesis'. Below it, text explains that the design example supports generation, simulation, and Quartus compilation flows. For the 'MIPI DSI-2 Rx+Tx Simulation Design Example', only the 'Simulation' option is available. For the 'MIPI DSI-2 TX Synthesis Design Example', only the 'Synthesis' option is available.
- Design Example Files:** Checkboxes for 'Synthesis' (checked) and 'Simulation' (unchecked). A 'Select Simulation Variant:' dropdown is set to 'Fast Simulation'. Text below explains that the design example simulation supports two options: 'Fast' and 'Full'. 'Fast simulation' has reduced runtime and is for simulating MIPI DSI-2 behavior. 'Full simulation' has a longer runtime and is for simulating the full MIPI DSI-2 and MIPI D-PHY system behavior.
- Target Development Kit:** A 'Select Board:' dropdown is set to 'No Development Kit'. Text below states that the design example supports generation, simulation, and Quartus compilation flows for any selected device. Hardware support is provided through a selected development kit. To exclude hardware aspects, 'None' should be selected from the dropdown.
- Target Device:** 'Device Selected: Family: Agilex 5 Device A5ED065BB32AE6SR0'. A 'Change Target Device' checkbox is present.

Table 14. Design Example Tab Parameter Description

Parameter	Values	Description
Select Design	MIPI DSI-2 TX Synthesis MIPI DSI-2 Rx+Tx Simulation	Select which design example is required. MIPI DSI-2 TX Synthesis provides a synthesizable system including MIPI DSI-2 TX and MIPI D-PHY IPs along with supporting system to demonstrate the correct connections to the IP. This can be used as the basis for a MIPI DSI-2 output subsystem in a user design. MIPI DSI-2 Rx+Tx Simulation provides a simulation system and all supporting scripts and models required to simulate the IP in a system.
Select Simulation Variant	Fast Simulation Full Simulation	When a simulation design example is generated, this option selects the system simulation variant. The fast simulation variant includes: - Video data source model - MIPI DSI-2 TX IP - Loopback at PPI interface - MIPI DSI-2 RX model - Video data checker

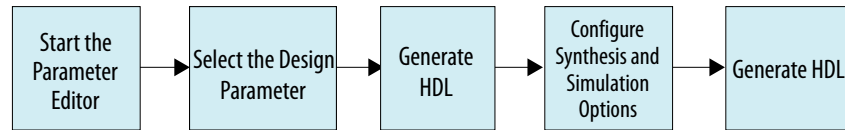
continued...

Parameter	Values	Description
		The full variant adds the MIPI D-PHY TX and MIPI D-PHY RX IPs, and performs the loopback at the MIPI serial interface. The full variant takes considerably longer to simulate.

3.3. Generating HDL for Synthesis and Simulation

Perform these steps to generate HDL for synthesis and simulation.

Figure 11. Steps to generate HDL for Synthesis and Simulation



1. Click **Generate HDL**. The **Generate** window appears as shown below.

Figure 12. HDL Generation for Synthesis and Simulation

Synthesis

Synthesis files are used to compile the system in a Quartus Prime project.

Create HDL design files for synthesis: Verilog

☐ Create timing and resource estimates for each IP in your system to be used with third-party EDA synthesis tools.

☐ Create block symbol file (.bsf)

☐ IP-XACT

☒ Generate IP Core Documentation

Simulation

The simulation model contains generated HDL files for the simulator, and may include simulation-only features.

Simulation scripts for this component will be generated in a vendor-specific sub-directory in the specified output directory.

Follow the guidance in the generated simulation scripts about how to structure your design's simulation scripts and how to use the *ip-setup-simulation* and *ip-make-simscript* command-line utilities to compile all of the files needed for simulating all of the IP in your design.

Create simulation model: None

Select simulation flow for specific simulators:

QuestaSim flow selection: QuestaSim

Select the simulators for which simulation scripts will be generated. If no simulators are selected, simulation scripts will be generated for all simulators.

☐ QuestaSim

☐ VCS(3-step)

☐ VCS(2-step, to be deprecated)

☐ Riviera-PRO

☐ Xcelium

Output Directory

☐ Clear output directories for selected generation targets.

Parallel IP Generation

If you select this option, Platform Designer performs IP generation with the number of processors defined in the Intel Quartus Prime parallel compilation settings (Assignments->Settings->Compilation Process Settings).

☒ Use multiple processors for faster IP generation (when available).

System Verilog Wrapper

☐ Create a System Verilog top-level wrapper when generating Verilog synthesis or simulation generation targets.

Generate Cancel

2. Configure the synthesis and simulation options.
3. Click **Generate** to complete the IP generation process.

3.3.1. Generated File Structure

The Quartus Prime Pro Edition software generates the following IP core output file structure:

Figure 13. Directory Structure

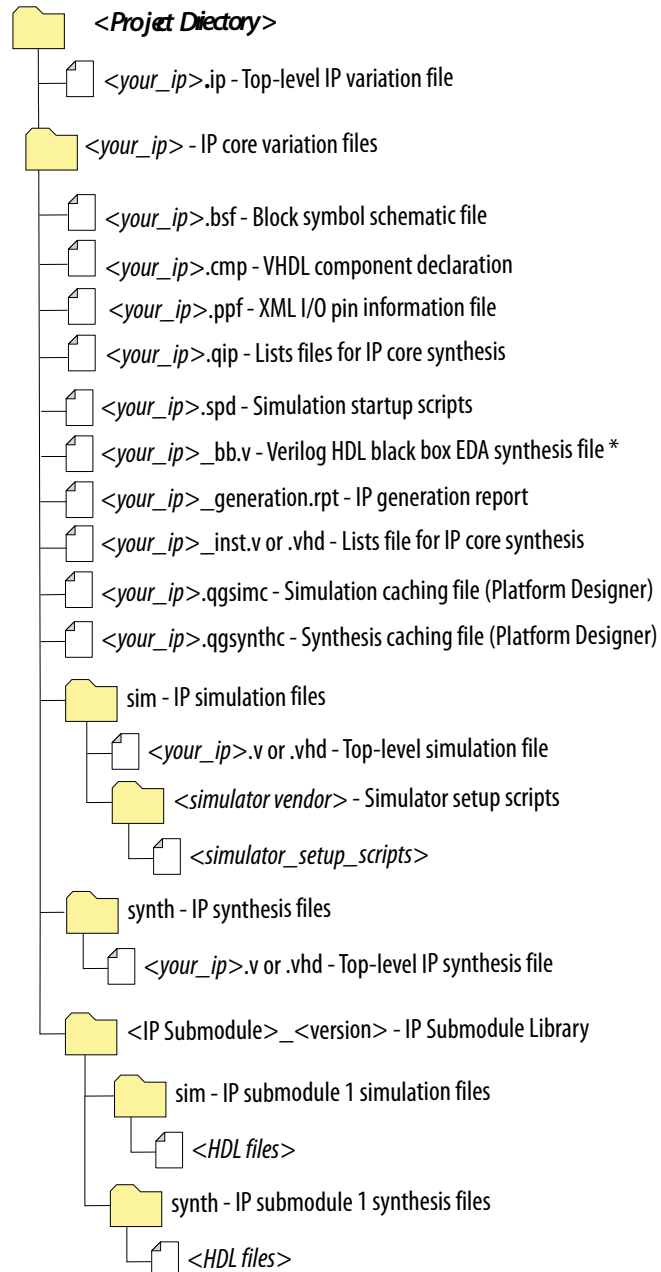
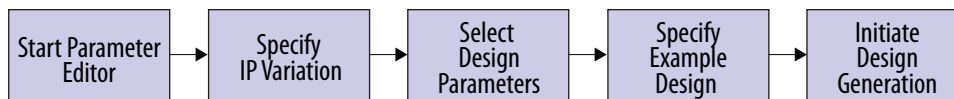


Table 15. Directory Structure and Files

Directory/File	Description
<my_ip>.cmp	The VHDL Component Declaration (.cmp) file is a text file that contains local generic and port definitions that you can use in VHDL design files.
<my_ip>_generation.rpt	IP or Platform Designer generation log file. A summary of the messages during IP generation.
<my_ip>.qggsimc	Simulation caching file that compares the .qsys and .ip files with the current parameterization of the Platform Designer system and IP. This comparison determines if Platform Designer can skip regeneration of the HDL.
<my_ip>.qggsynth	Synthesis caching file that compares the .qsys and .ip files with the current parameterization of the Platform Designer system and IP. This comparison determines if Platform Designer can skip regeneration of the HDL.
<my_ip>.qip	Contains all the required information about the IP component to integrate and compile the IP component in the Quartus Prime software.
<my_ip>.csv	Contains information about the upgrade status of the IP component.
<my_ip>.bsf	A Block Symbol File (.bsf) representation of the IP variation for use in Block Diagram Files (.bdf).
<my_ip>.spd	Required input file for ip-make-simscript to generate simulation scripts for supported simulators. The .spd file contains a list of files generated for simulation, along with the information about the memories that you can initialize.
<my_ip>_bb.v	Use the Verilog black box (_bb.v) file as an empty module declaration for use as a black box.
<my_ip>_inst.v or <my_ip>_inst.vhd	HDL example instantiation template. Copy and paste the contents of this file into your HDL file to instantiate the IP variation.
<my_ip>.v or <my_ip>.vhd	HDL files that instantiate each submodule or child IP core for synthesis or simulation.
sim/mentor/	Contains a Questa* – Altera® FPGA Edition script msim_setup.tcl to set up and run a simulation.
sim/synopsys/vcsmx/	Contains a shell script vcsmx_setup.sh and synopsys_sim.setup file to set up and run a VCS* MX simulation.
sim/xcelium/	Contains a shell script xcelium_setup.sh and other setup files to set up and run a Xcelium* simulation.
sim/common/	Contains a set of Tcl files, <simulator>_files.tcl, which provide all design related simulation information required by a corresponding simulation script. The Tcl file contains designs from current system-level hierarchy, and references to sub-systems and IP components.
<IP_submodule>/	For each generated IP submodule directory, Platform Designer generates /synth and /sim sub-directories under IP-specific directories.

3.4. Generating the Design Example

Figure 14. Procedure to Generate Design Example



1. In the IP Catalog, locate and select **Library > Interface Protocols > Audio & Video > MIPI DSI-2 IP**. The **New IP Variation** window appears.
2. Specify a top-level name *<your_ip>* for your custom IP variation. The parameter editor saves the IP variation settings in a file named *<your_ip>.ip*.
3. Click **OK**. The parameter editor appears.
4. Configure the parameters for your design in the **IP**, **Video Timing**, and **Design Example** tabs.
5. Click the **Generate Example Design** button. The **Select Example Design Directory** window appears.
6. Once the design example is generated, click the **Launch Example Design in Quartus**.
7. Click **OK** to launch the generated design example in Quartus Prime.

3.4.1. Generated Directory Structure and Files

Table 16. Generated Files in the quartus Directory and File Descriptions

The `quartus` directory contains the Quartus Prime project files, including compilation reports:

File	Description
<code>dsi2_<variant>_sys.qpf</code>	Quartus Prime project file.
<code>dsi2_<variant>_sys.qsf</code>	Quartus Prime .qsf assignment file.

Table 17. Generated Files in the rtl Directory and File Descriptions for Synthesis and Full Simulations

The `rtl` directory contains all synthesizable components, including the system interconnect and Platform Designer generated IP:

Directory or File	Description
<code>dsi2_dphy_sys.qsys</code>	Platform Designer file that contains MIPI DSI-2 and DPHY system definition.
<code>dsi2_dphy_sys/</code>	Directory that contains Platform Designer generated files.
<code>ip/dsi2_dphy_sys</code>	Directory that contains the Platform Designer generated IP files.
<code>.../dsi2_dphy_sys_dsi2_tx.ip</code> <code>.../dsi2_dphy_sys_dsi2_tx/</code>	IP definition file for MIPI DSI-2.
<code>.../dsi2_dphy_sys_mipi_dphy.ip</code> <code>.../dsi2_dphy_sys_mipi_dphy/</code>	IP definition file for MIPI D-PHY IP (unused in the fast simulation variant).
<code>.../dsi2_dphy_sys_reset_release_0.ip</code> <code>.../dsi2_dphy_sys_reset_release_0/</code>	IP definition file for Reset Release.
<i>continued...</i>	

Directory or File	Description
.../dsi2_dphy_sys_dsi2_tx_axi4s_clk_bridge.ip .../dsi2_sys_dphy_dsi2_tx_axi4s_clk_bridge/	IP definition file for Clock Bridge for MIPI DSI-2 AXI-S Clock.
.../ dsi2_dphy_sys_dsi2_tx_axi4s_reset_bridge.ip .../dsi2_dphy_sys_dsi2_tx_axi4s_reset_bridge/	IP definition file for Reset Bridge for MIPI DSI-2 AXI-S Clock.
.../ dsi2_dphy_sys_mipi_dphy_link1_link_core_clock_bridge.ip .../ dsi2_dphy_sys_mipi_dphy_link1_link_core_clock_bridge/	IP definition file for Clock Bridge for DPHY Clock.
.../ dsi2_dphy_sys_mipi_dphy_link1_link_core_reset_bridge.ip .../ dsi2_dphy_sys_mipi_dphy_link1_link_core_reset_bridge/	IP definition file for Reset Bridge for DPHY Clock.

Table 18. Generated Files in ed_sim Directory and File Descriptions

The ed_sim directory contains all simulation files and scripts:

Directory / File	Description
dsi2_ed_sim_tb.sv	Top-level testbench file.
dut_wrapper.sv	Wrapper around MIPI CSI-2 TX, RX BFM and MIPI Loopback. In the fast simulation variant, the PPI interface from the MIPI DSI-2 TX IP is looped back into the MIPI DSI-2 RX BFM. In the full simulation variant, the MIPI DSI-2 TX PPI is connected to MIPI D-PHY TX IP, then looped back to MIPI D-PHY RX IP and into the MIPI DSI-2 RX BFM.
cfg_pkg.sv	SystemVerilog package containing generated IP parameters.
sim.spd	File used by Quartus Prime ip-make-simscript.
mipi_*.sv	Simulation-only encrypted MIPI DSI-2 receiver bus functional model.
hdl/	Contains Encrypted Verification IP files.
sim/aldec/	Contains Riviera-PRO* simulator script.
run_rivierapro_setup.tcl	Script that runs Riviera-PRO simulation.
rivierapro_setup.tcl	Setup file for Riviera-PRO simulation.
sim/mentor/	Contains Mentor simulator script.
run_msim_setup.tcl	Script that runs QuestaSIM* simulation.
msim_setup.tcl	Setup file for QuestaSIM simulation.
sim/synopsys/vcsmx/	Contains Synopsys VCS MX simulator script.
sim.sh	Script that runs VCS MX simulation.
vcsmx_setup.sh	Setup script for VCS MX simulation.
continued...	

Directory / File	Description
synopsys_sim_setup.sh	Setup file for VCS MX simulation.
sim/xcelium/	Contains Xcelium simulator script.
xcelium_setup.sh	Script that runs Xcelium simulation.

3.5. Compiling the Design Example

After generating the design example, follow these steps to start compilation:

1. In the Quartus Prime Pro Edition software, open the Quartus Prime Pro Edition project `<project_directory>/quartus/dsi2_dphy_sys.qpf`.
2. On the **Processing** menu, click **Start Compilation**.

4. Integrating the MIPI DSI-2 IP With Your Application

This section provides information on how to integrate the IP into your system, with information on interfaces, clocks, resets and control.

4.1. High-Level Interface Types

4.1.1. Transmitter Input Video Interface Signals

This interface carries video data into the MIPI DSI-2 IP. It complies with the Altera Streaming Video Protocol (full variant) which transports pixel data via an AXI4-Stream-compliant interface. For more information on this protocol and rules for pixel formatting and signaling, refer to the *Altera Streaming Video Protocol Specification*.

The video interface signal widths can be calculated with the following formula:

$$P = \max\left(2, \text{ceil}\left(\frac{\text{bits_per_color_plane} \times \text{number_of_color_planes}}{8}\right)\right) \times \text{pixels_in_parallel} \times 8$$

$$Q = \text{ceil}\left(\frac{P}{8}\right)$$

The following table shows some examples on the bit widths values:

Table 19. Examples for Video Interface Bit Widths

Data Type	Bits per Color Plane	Number of Color Planes
RGB 16bit (RGB565)	6	3
RGB 24bit (RGB888)	8	3
YCbCr 12bit (4:2:0)	10	1
YCbCr 16bit (4:2:2)	8	2

Table 20. Transmitter AXI-4 Stream Input Video Interface Signals

Signal Name	Width	Direction	Description
axi4s_vid_in_tdata	P	Input	AXI4-Stream data.
axi4s_vid_in_tvalid	1	Input	AXI4-Stream data valid.
axi4s_vid_in_tuser	Q	Input	Bit 0: Start of video frame • 0 = Not start of field • 1 = Start of field Bit 1: Meta or video packet

continued...

Signal Name	Width	Direction	Description
			0 = Video packet 1 = Meta packet - Bits Q-1:2 = Unused
axi4s_vid_in_tlast	1	Input	AXI4-Stream end of packet.
axi4s_vid_in_tready	1	Output	AXI4-Stream data ready.

4.1.2. Transmitter PHY-Protocol Interface (PPI) Signals

The interface between the MIPI D-PHY IP and the MIPI DSI-2 IP uses the PHY-Protocol interface defined in the MIPI D-PHY and MIPI DSI-2 standards. This carries the outgoing MIPI data stream from the MIPI DSI-2 transmitter to be processed by the MIPI D-PHY IP.

Table 21. Transmitter AXI-4 Stream Input Video Interface Signals

Signal Name	Width	Direction on MIPI DSI-2 IP	Description
Clock Lane High Speed Transmit: <ppi_width> = 16			
tx_word_clk_hs_ck	1	Input	Refer to MIPI D-PHY specification version 2.5: Figures 92-94 Example High-Speed Transmission from the Master Side: 8-/16-/32-Bit Bus Width.
tx_data_width_hs_ck	2	Output	
tx_data_hs_ck	ppi_width	Output	
tx_valid_hs_ck	ppi_width/8	Output	
tx_data_transfer_en_hs_ck	1	Output	
tx_request_hs_ck	1	Output	Refer to MIPI D-PHY specification version 2.5: Figure 90, Example High Speed Clock Enable, LP Mode. Figure 121 High Speed Transmission and Reception with HS-Idle Function.
tx_ready_hs_ck	1	Input	
tx_eq_active_hs_ck	1	Output	Refer to MIPI D-PHY specification version 2.5 Table 50.
tx_eq_level_hs_ck	1	Output	
tx_skew_cal_hs_ck	1	Output	Skew calibration. Implementation-specific how the system responds to this condition. Refer to MIPI D-PHY specification version 2.5: Figure 119. Skew Calibration, LP Mode.
tx_alternate_cal_hs_ck	1	Output	
Data Lane High Speed Transmit: <lane>=0-3; <ppi_width>=16			
tx_word_clk_hs_d<lane>	1	Input	Refer to MIPI D-PHY specification version 2.5: Figures 92-94 Example High-Speed Transmission from the Master Side: 8-/16-/32-Bit Bus Width.
tx_data_width_hs_d<lane>	2	Output	
tx_data_hs_d<lane>	ppi_width	Output	
tx_valid_hs_d<lane>	ppi_width/8	Output	
tx_data_transfer_en_hs_d<lane>	1	Output	
tx_request_hs_d<lane>	1	Output	Refer to MIPI D-PHY specification version 2.5: Figure 90, Example High Speed Clock Enable, LP Mode.
tx_ready_hs_d<lane>	1	Input	
continued...			

continued...

Signal Name	Width	Direction on MIPI DSI-2 IP	Description
			Figure 121 High Speed Transmission and Reception with HS-Idle Function.
tx_eq_active_hs_d<lane>	1	Output	Refer to MIPI D-PHY specification version 2.5 Table 50.
tx_eq_level_hs_d<lane>	1	Output	
tx_skew_cal_hs_d<lane>	1	Output	Calibration. Implementation-specific how the system responds to this condition. Refer to MIPI D-PHY specification version 2.5: Figure 119. Skew Calibration, LP Mode.
tx_alternate_cal_hs_d<lane>	1	Output	
Clock Lane Escape Mode Transmit: (LPDT= CCI Control Data Transmission, ULPS = Sleep Mode)			
tx_clk_esc_ck	1	Input	Escape mode signals – Refer to MIPI D-PHY specification version 2.5 Table 50.
tx_request_esc_ck	1	Output	
tx_request_type_esc_ck	4	Output	
tx_lpdt_esc_ck	1	Output	
tx_ulps_exit_ck	1	Output	
tx_ulps_esc_ck	1	Output	
tx_trigger_esc_ck	4	Output	
tx_data_esc_ck	8	Output	
tx_valid_esc_ck	1	Output	
tx_ready_esc_ck	1	Input	
Data Lane Escape Mode Transmit: (LPDT= CCI Control Data Transmission, ULPS = Sleep Mode). <lane>=0-3			
tx_clk_esc_d<lane>	1	Input	Escape mode signals – Refer to MIPI D-PHY specification version 2.5 Table 50.
tx_request_esc_d<lane>	1	Output	
tx_request_type_esc_d<lane>	4	Output	
tx_lpdt_esc_d<lane>	1	Output	
tx_ulps_exit_d<lane>	1	Output	
tx_ulps_esc_d<lane>	1	Output	
tx_trigger_esc_d<lane>	4	Output	
tx_data_esc_d<lane>	8	Output	
tx_valid_esc_d<lane>	1	Output	
tx_ready_esc_d<lane>	1	Input	
Clock Lane Control			
turn_request_ck	1	Output	Clock lane direction signals – Refer to MIPI D-PHY specification version 2.5 Table 50.
direction_ck	1	Input	
turn_disable_ck	1	Output	
continued...			

Signal Name	Width	Direction on MIPI DSI-2 IP	Description
force_rx_mode_ck	1	Output	Refer to MIPI D-PHY specification version 2.5: Figure 86. Master PHY Enable, LP Mode.
force_tx_stop_mode_ck	1	Output	
stop_state_ck	1	Input	
enable_ck	1	Output	
alp_mode_ck	1	Output	Alternate Low Power Mode Selection – Refer to MIPI D-PHY specification version 2.5 Table 50.
tx_ulps_clk_ck	1	Output	Refer to MIPI D-PHY specification version 2.5: Figure 114. Example Clock Lane ULPS Entry and Exit, LP Mode.
rx_ulps_clk_not_ck	1	Input	
ulps_active_not_ck	1	Input	
tx_hsidle_clk_hs_ck	1	Output	Refer to MIPI D-PHY specification version 2.5: Figure 121. HS Transmission and Reception with HS-IDLE Function.
tx_hsidle_clk_ready_hs_ck	1	Input	
Data Lane Control: <lane>=0--3			
turn_request_d<lane>	1	Output	Data lane direction signals – Refer to MIPI D-PHY specification version 2.5 Table 50.
direction_d<lane>	1	Input	
turn_disable_d<lane>	1	Output	
force_rx_mode_d<lane>	1	Output	Refer to MIPI D-PHY specification version 2.5: Figure 86. Master PHY Enable, LP Mode.
force_tx_stop_mode_d<lane>	1	Output	
stop_state_d<lane>	1	Input	
enable_d<lane>	1	Output	
alp_mode_d<lane>	1	Output	Alternate Low Power Mode Selection – Refer to MIPI D-PHY specification version 2.5 Table 50.
tx_ulps_clk_d<lane>	1	Output	Refer to MIPI D-PHY specification version 2.5: Figure 114. Example Clock Lane ULPS Entry and Exit, LP Mode.
rx_ulps_clk_not_d<lane>	1	Input	
ulps_active_not_d<lane>	1	Input	
tx_hsidle_clk_hs_d<lane>	1	Output	Refer to MIPI D-PHY specification version 2.5: Figure 121. HS Transmission and Reception with HS-IDLE Function.
tx_hsidle_clk_ready_hs_d<lane>	1	Input	

4.1.3. Avalon Memory-Mapped Interface Control Signals

The Avalon Memory-Mapped agent control interface provides access to the control and status registers in the MIPI DSI-2 IP and CVO subcore to a host processor

This interface operates on the same clock domain as the Altera Streaming Video input on the MIPI DSI-2 IP.

For details on the register map, see [Appendix B: Registers](#).

Table 22. Avalon Memory-Mapped Interface Control Signals

Signal Name	Width	Direction	Description
control_address	10	Input	Standard Avalon memory-mapped interface, supporting transfers in burst mode with full handshaking. Write transfers always have a wait time of 0 cycle while read transfers have a wait time of 1 cycle. The interface should be accessed using word addressing in the Platform Designer flow. For example, an address of 4 in the Nios® V software selects the address of 1 in the agent.
control_write	1	Input	
control_byteenable	4	Input	
control_writedata	32	Input	
control_read	1	Input	
control_readdata	32	Output	
control_readdatavalid	1	Output	
control_waitrequest	1	Output	Active high interrupt signal.
control_irq	1	Output	

4.1.4. Additional Control Signals

This additional signal operates on the same clock domain as the Altera Streaming Video input on the MIPI DSI-2 IP.

Table 23. Additional Control Signals

Signal Name	Width	Direction	Description
frame_start	1	Input	Frame start indicator. This signal can be used to restart the timing raster generated by the Video Timing Generator in the CVO. For further details on how to use this, see the <i>Video Timing Generator</i> section in the <i>Altera VVP Suite User Guide</i> .

Related Information

[Video Timing Generator IP](#)

4.2. Dependent/Supporting IPs

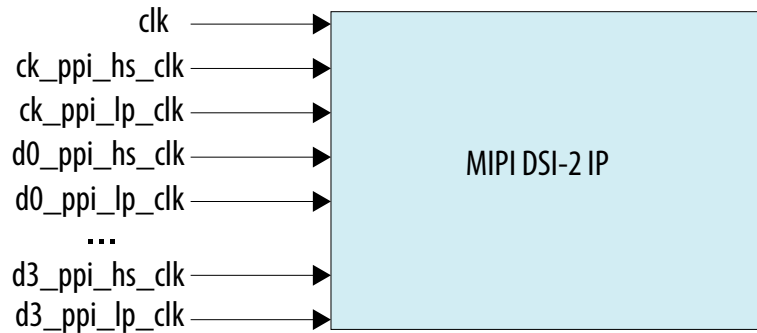
The video input interface is designed to receive video from logic that supports the Altera Streaming Video Protocol (full variant). The Altera Video and Vision Processing (VVP) IP suite provides a range of video processing IP using this protocol that can be used in a wide range of video systems.

The MIPI PPI interfaces to the Altera MIPI D-PHY IP in Agilex 3 and Agilex 5 FPGAs. This performs the low-level interfacing to external MIPI D-PHY receivers.

4.3. Implementing Required Clocking

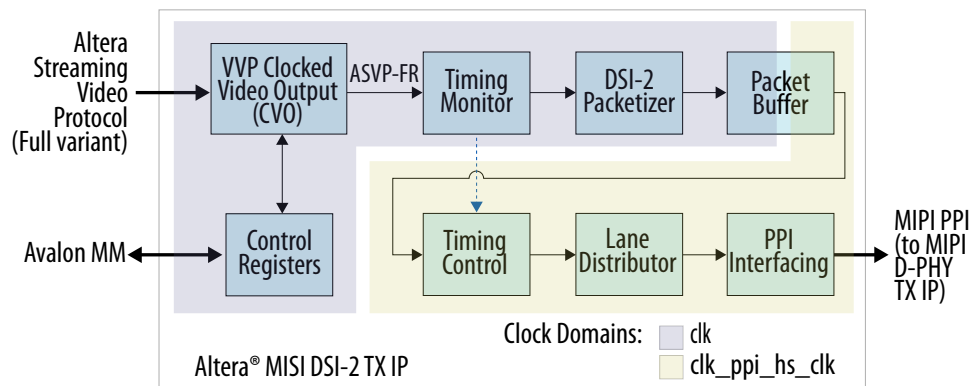
The MIPI DSI-2 IP clock input signals are shown in the following diagram. There are no clock outputs from the IP.

Figure 15. IP Clocks



The clock domains within the IP are as below:

Figure 16. IP Clock Domain



4.3.1. MIPI PPI Clocks

The MIPI PPI connection between the MIPI DSI-2 and MIPI D-PHY IPs includes multiple clock signals from the MIPI D-PHY IP. The lane distribution and packet transmission within the MIPI DSI-2 IP is always performed in the MIPI D-PHY link clock domain.

Guidance on the required connections for each clock is in the following table, where `<lane>` indicates each of the active lane for the configured MIPI connection and `<x>` indicates the appropriate `LINK` number for this MIPI channel on the MIPI D-PHY IP. These signals must be connected directly between the MIPI DSI-2 and MIPI D-PHY IPs.

The frequency of these clocks is determined by the configuration of the MIPI D-PHY IP and the external MIPI interface.

Table 24. PPI Clock Connections

Signal Name	Direction	Description
ck_ppi_hs_clk	Input	High-speed clock on MIPI clock lane. Connect to LINK<x>_ck_ppi_hs_clk on MIPI D-PHY.
ck_ppi_tx_lp_clk	Input	Low power clock on MIPI clock lane. Connect to LINK<x>_ck_ppi_lp_clk on MIPI D-PHY.
d<lane>_ppi_hs_clk	Input	High-speed clock on MIPI data lanes. Connect to LINK<x>_d<lane>_ppi_hs_clk on MIPI D-PHY.
d<lane>_ppi_tx_lp_clk	Input	Low power clock on MIPI data lanes. Connect to LINK<x>_d<lane>_ppi_lp_clk on MIPI D-PHY.

4.3.2. Video Input Clock

The Altera Streaming Video input interface `axi4s_vid_in` operates on the `clk` clock domain.

In MIPI DSI-2 IP version 1.0.0, the `clk` clock input must be connected to the `LINK<x>_link_core_clk` on the MIPI D-PHY IP. In this configuration, the IP processes the incoming video and produces the MIPI output using a single clock domain throughout and without any clock domain crossings. This results in deterministic processing latency especially with fixed latency mode enabled and reduced resource utilization.

In this configuration, the video source to the IP must also be running on the MIPI link clock. In systems where the video is generated in another domain, it can be transferred onto this domain before connecting to the MIPI DSI-2 IP using the Video Streaming FIFO IP in dual clock configuration. The synthesis design example exports the link clock as `link_core_clk_bridge_out_clk` and associated reset, which the user logic should use to drive the video into the MIPI DSI-2 video input.

From MIPI DSI-2 IP version 2.0.0 onwards, an additional user parameter `CLOCKS_ARE_SAME` allows the user to select whether the `clk` clock input is common with the MIPI PPI output clock or is independent.

When the `CLOCKS_ARE_SAME` parameter is enabled, the clock architecture is the same as that in version 1.0.0. When the parameter is disabled, the input clock is independent of the MIPI PPI and can be driven by any suitable clock. This allows a simpler system clock architecture with freedom to run the whole video pipeline in the FPGA on a single clock domain. When the input clock is independent, it is important to ensure that the clock provides video data at the input to the MIPI DSI-2 IP fast enough to ensure that the MIPI output is always satisfied and cannot underflow.

The AXI clock may operate at any frequency up to the maximum clock rate supported by the FPGA.

Related Information

[Device Speed Grade Support](#) on page 6

4.3.3. Control Interface Clock

The Avalon Memory-Mapped control interface also operates on the `clk` domain. The `CLOCKS_ARE_SAME` parameter operates on this interface in the same manner as the video input interface. When this parameter is enabled, the interface operates on the MIPI link clock. When disabled, the clock is independent of the MIPI output.

4.4. Implementing Required Resets

The MIPI DSI-2 IP has a single reset input signal, `reset`.

Table 25. Reset Signals

Signal Name	Width	Direction	Description
<code>reset</code>	1	Input	This signal is used to reset all logic within the IP. It is an asynchronous reset, active high, and must be asserted for at least 256 clock cycles after <code>clk</code> is active.

4.5. Connecting the Video Input Interface

The video input interface to the MIPI DSI-2 IP uses the Altera Streaming Video Protocol (full variant). This is based on the AMBA AXI4-Stream interface, carrying video and control packets describing the video structure.

The interface is synchronous to the `clk` clock. Flow control is implemented using the `TVALID` and `TREADY` signals.

Table 26. Video Interface Signals

Signal Name	Width	Direction	Description
<code>axi4s_vid_in_tdata</code>	<code>TDATA_WIDTH</code>	Input	This carries the pixel and control packet data. The width follows the interface configuration as described in the Protocol specification and also displayed on the IP parameter editor.
<code>axi4s_vid_in_tvalid</code>	1	Input	Indicates that the current cycle carries valid data.
<code>axi4s_vid_in_tready</code>	1	Output	Asserted by the IP when the current cycle can be processed. If this signal is deasserted, the contents of the interface are not captured on this cycle.
<code>axi4s_vid_in_tlast</code>	1	Input	Indicates the last cycle in a control or video packet.
<code>axi4s_vid_in_tuser</code>	$\text{TDATA_WIDTH}/8$	Input	- Bit [0] is asserted for the first cycle in a video frame. - Bit [1] is asserted for the first cycle in a control (Image Information) packet.

For more details on the Altera Streaming Video protocol and pixel formatting, refer to *Altera FPGA Streaming Video Protocol Specification*.

Related Information

[Altera FPGA Streaming Video Protocol Specification](#)

4.6. Connecting the MIPI PHY-Protocol Interface

The MIPI PHY-Protocol Interface (PPI) connects the data stream from the MIPI DSI-2 IP to the MIPI D-PHY IP. The signal functions and clock domains are as defined in the MIPI D-PHY specification v2.5 Annex A.

Each of the MIPI clock and data lane has an independent set of signals that are connected between the MIPI DSI-2 IP and the ports for the appropriate link for that channel on the MIPI D-PHY IP. All PPI signals should be directly connected to the corresponding signal on the MIPI D-PHY IP. In general, there should not be any other connections to these signals. Ensure that connections do not interconnect signals between the different lanes.

For more details on the MIPI D-PHY PPI signals, refer to the *MIPI D-PHY IP User Guide: Agilex 3 and Agilex 5 FPGAs*.

Related Information

[MIPI D-PHY IP User Guide: Agilex 3 and Agilex 5 FPGAs](#)

4.7. Connecting the Control Interface

The optional control interface is a standard Avalon memory-mapped interface used to access control and status registers.

If this interface is enabled, these signals should be connected to a control host directly or via interconnect logic, such as that generated by Platform Designer. The register map for this interface is in [Appendix B - Registers](#).

If this interface is not enabled, the status registers in the MIPI DSI-2 IP are not available and CVO operation follows the fixed parameter configuration set in the Parameter Editor.

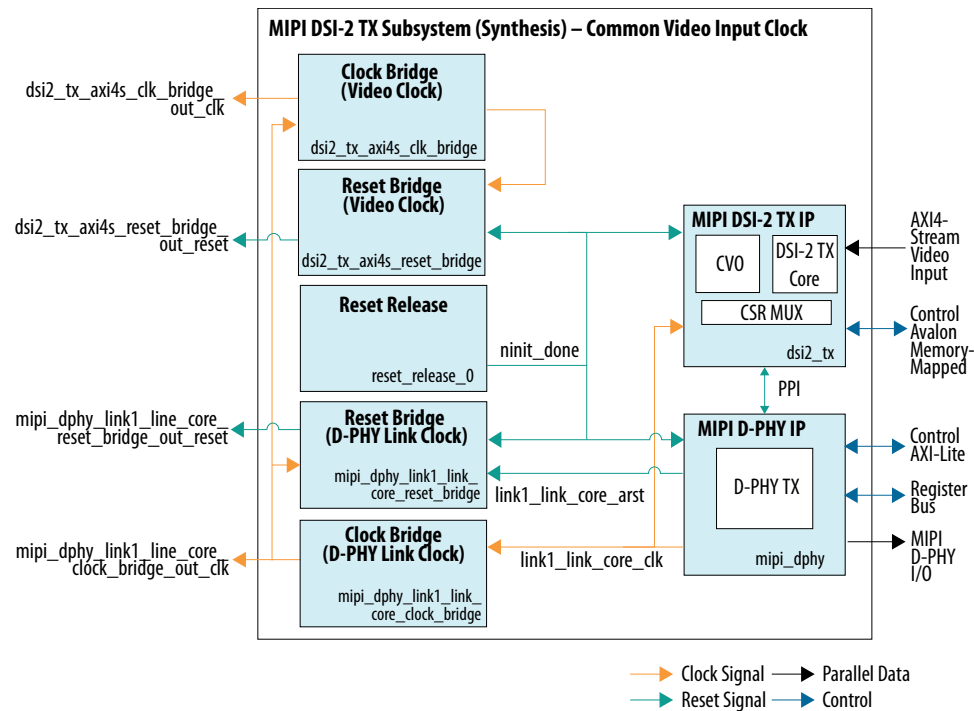
For more details on the signaling on this interface, refer to the *Avalon memory-mapped interface Transfers* in the *Avalon Interface Specifications*.

Related Information

[Avalon Memory-Mapped Interfaces Transfers](#)

5. Synthesizing the IP

Figure 17. MIPI DSI-2 TX Subsystem (Synthesis) – Common Video Input Clock Block Diagram



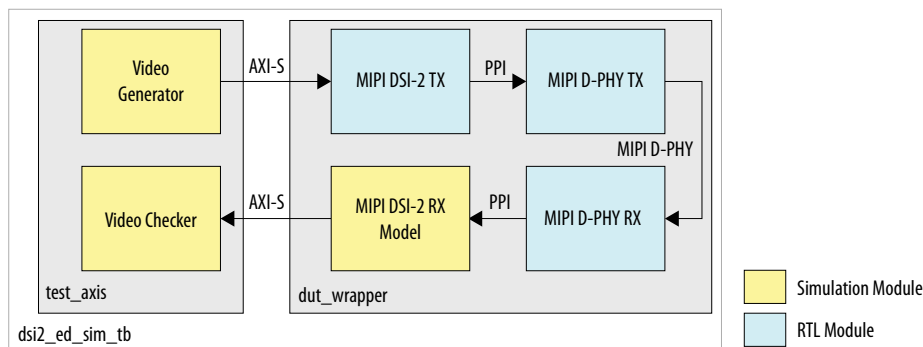
This diagram illustrates the MIPI DSI-2 TX synthesis design example, where both video processing and the MIPI D-PHY IP share a common video input clock. Clock Bridges and Reset Bridges synchronize and distribute the shared clock and reset signals to both domains. Video enters through an AXI4-Stream interface, is processed by the MIPI DSI-2 Tx IP, and transmitted via the MIPI D-PHY TX IP to external MIPI-compliant devices. Control and configuration are handled through Avalon memory-mapped interface and AXI-Lite interface. Using a shared clock simplifies timing management and ensures seamless synchronization between video and physical layer.

6. Simulating the IP

6.1. Design Example Simulation Features

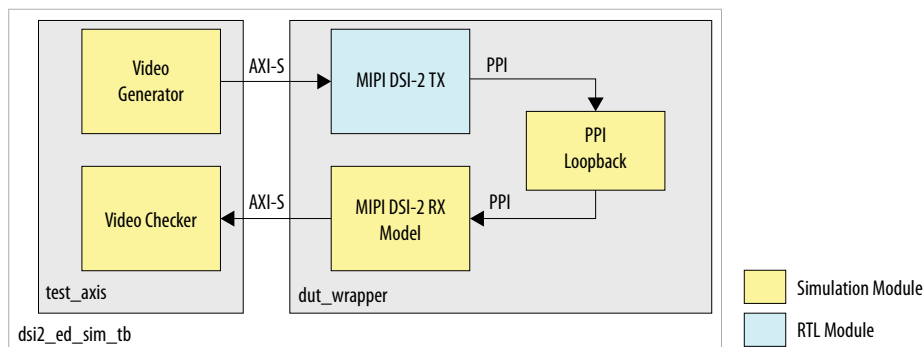
The simulation design example allows the operation of the MIPI DSI-2 IP to be observed and tested in a loopback system. The simulation produces video data and supplies it to the DSI TX IP and loops the output packet data back into a DSI RX model before comparing the recovered video data with the input data. Two variants of the system are provided, which are described below. In each case, the configuration and operation of the whole system is adapted to the user's configuration of the MIPI DSI-2 TX IP as the design example simulation is generated.

Figure 19. Full Simulation Block Diagram



The full simulation variant sends video through the MIPI DSI-2 TX and MIPI D-PHY TX and loops back the MIPI serial signals to the MIPI D-PHY RX IP. It then recovers and checks the video data. This gives an accurate view of the operation of the MIPI subsystem to the MIPI D-PHY external I/O level.

Figure 20. Fast Simulation Block Diagram



The fast simulation variant ignores the MIPI D-PHY IPs and performs loopback at the PPI level. A simple PPI loopback module replaces the MIPI D-PHY IPs and adapts from the PPI TX to PPI RX signaling while mimicking the MIPI D-PHY flow control. This allows the simulation to run much faster and generally provides a sufficient view of the way that the MIPI DSI-2 TX IP functions. The operation of the simulation is otherwise identical to the full simulation variant.

6.2. Design Example Components

Table 27. Design Example Components

Module	Description
dsi2_ed_sim_tb	The top-level testbench wrapper, which instantiates the test logic and the DUT.
test_axis	This wraps the test logic that triggers the production of test data and checks the output data.
dut_wrapper	This contains the system under test, with the MIPI DSI-2 TX IP and the other modules required to create the system being tested.
Video Generator	This produces data that takes the form of video pixel data on the Altera Video Streaming Protocol (full variant) interface.
MIPI DSI-2 TX	MIPI DSI-2 TX IP in the configuration.
MIPI D-PHY TX	MIPI D-PHY transmitter IP in a configuration compatible with the MIPI DSI-2 TX IP.
PPI Loopback	A simulation adapter that allows the TX PPI signals to be directly connected to the RX PPI signals for a simple loopback connection without the MIPI D-PHY IP in the fast simulation variant.
MIPI D-PHY RX	MIPI D-PHY receiver IP in a configuration compatible with the MIPI DSI-2 TX IP.
MIPI DSI-2 RX Model	This is a behavioral model of a MIPI DSI-2 receiver, used to recover the video data from the RX PPI for checking.
Video Checker	Compares the video pixel data against the data transmitted. If any mismatch is detected, simulation errors are produced.

6.3. Supported Simulation EDA Tools

The MIPI DSI-2 simulation design example supports the following simulators:

- Questa – Altera FPGA Edition
- Siemens* EDA QuestaSIM
- VCS
- Xcelium

Refer to *Supported Simulators* in the *Quartus Prime Pro Edition User Guide: Third-party Simulation*.

Related Information

[Supported Simulators](#)

6.4. Simulating the Design Example

You can simulate the design by running a simulation script from the command prompt.

Note: The vertical active size is fixed to 96 lines. As a result, VTOTAL must be configured to (V1S_VEND + 96) when generating the simulation for it to proceed correctly. This only relates to the simulation design example and does not affect synthesis or simulation of the IP in user systems.

1. At the command prompt, change the working directory of your preferred simulator based on the table below.
2. Run the simulation scripts for the supported simulator.

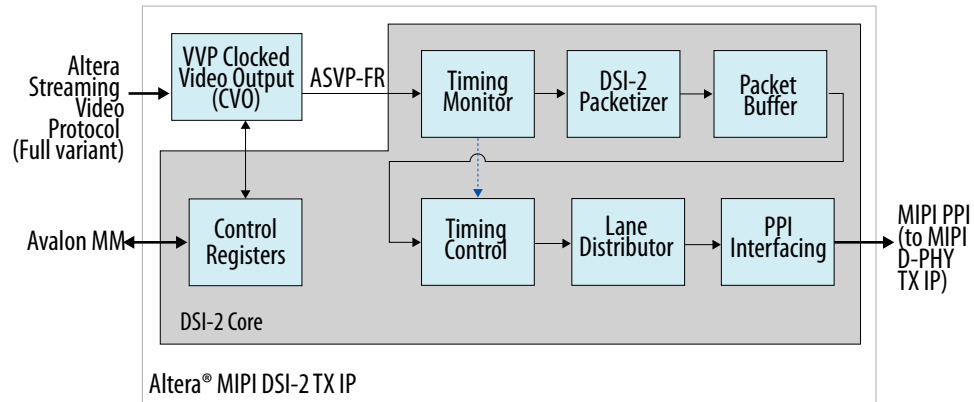
Simulator	Working Directory	Command
QuestaSIM	ed_sim/sim/mentor	<code>vsim -c -do run_msim_setup.tcl</code>
Riviera-PRO	ed_sim/sim/aldec	<code>vsim -c -do run_rivierapro_setup.tcl</code>
VCS MX	ed_sim/sim/synopsys/vcsmx	<code>source sim.sh</code>
Xcelium	ed_sim/sim/xcelium	<code>source xcelium_setup.sh</code>

A successful simulation displays the following message:

```
# dsi2_ed_sim_tb: SUCCESS: Test has completed!  
# Simulation passed.
```

After successful completion, you can analyze the results.

7. Appendix A: Functional Description



The MIPI DSI-2 TX IP receives video from the Altera Streaming Video Protocol (full variant). The Clocked Video Output (CVO) module from the Altera Video & Vision Processing (VVP) IP suite forms a video raster using the configured timing and inserts the incoming video into this raster. The Altera Streaming Video Protocol (full raster) format is used to carry the video from the CVO into the MIPI DSI-2 core.

The MIPI DSI-2 IP processes the video data and video synchronization information, then converts them into packets as defined by the MIPI DSI/DSI-2 specification to prepare them for transmission via the Altera MIPI D-PHY IP.

The Timing Monitor and Timing Control modules are only included when fixed latency mode is enabled. This mode records the timing of the video and synchronization pulses on arrival from the CVO. The MIPI DSI-2 packets are retimed using this information to ensure maximum timing accuracy with minimal additional FPGA resources consumption. When this mode is disabled, the MIPI DSI-2 IP operates on a minimum latency basis and there may be some variation in inter-packet timing on the MIPI external interface. For example, sync timing may be offset horizontally relative to the video.

The Timing Monitor records timestamps from the video and synchronization pulses from the CVO for use in controlling packet playout in Timing Control.

The MIPI DSI-2 Packetizer packs the input pixels to MIPI DSI-2 packet payload data as defined in the MIPI DSI-2 specification. Packet headers and checksums are attached. Syncs are converted into MIPI DSI-2 synchronization packets. When ECC and CRC calculation is enabled, these are inserted into the relevant part of the packets. When disabled, these values are fixed to zero.

A Packet Buffer holds the MIPI DSI-2 packets. When fixed latency mode is disabled, the packet buffer stores the packet data until a complete packet is ready to send, to ensure contiguous packets are sent on the MIPI D-PHY interface. In fixed latency mode, this is ensured by the fixed latency timing logic in Timing Control.

Timing Control holds packets until a fixed period from the corresponding events at the Timing Monitor has expired before allowing each packet to be forwarded to ensure fixed latency through the packetizer and packet buffer.

The Lane Distributor spreads the packet data across the configured number of lanes, as defined by the MIPI DSI-2 specification.

The PPI Interfacing module handles the signaling of the data to the MIPI D-PHY IP via the MIPI PHY-Protocol Interface (PPI) defined in the MIPI D-PHY specification.

An optional Avalon memory-mapped control interface is presented, which allows access to the registers in both the MIPI DSI-2 core and CVO modules. The registers in the MIPI DSI-2 core only display the status information, while the registers in the CVO and Video Timing Generator allow for reconfiguration of the output video timing. When the control interface is disabled, the video timing is fixed according to the configuration in the Parameter Editor.

8. Appendix B: Registers

Table 28. Types of Register Access

Access	Definition
RO	Read only.
RW	Read and write.
RWC	Read, and write and clear. The user application writes 1 to the register bit(s) to invoke a defined instruction. The IP clears the bit(s) upon executing the instruction.

Observe the following guidelines when accessing the registers:

- Do not write to reserved or undefined registers.
- When writing to the registers, perform read-modify-write operation to ensure that reserved or undefined register bits are not overwritten.

8.1. MIPI DSI-2 IP Base Registers

The following 32-bit status registers are accessible when the Control and Status Registers (CSRs) option is enabled. Addresses are given as byte offsets from the base address of the MIPI DSI-2 instance. When accessed using 32-bit word addresses, these offsets should be divided by four.

Table 29. MIPI DSI-2 IP Base Register Map

Address	Register Name	Access	Default	Bits	Description
0x00	LANES	RO	0x02	3:0	Data lanes configured in IP.
0x04	PIXELS_IN_PARALLEL	RO	0x02	3:0	Pixels in parallel configured in IP.
0x08	DATA_TYPE	RO	0x3E	5:0	MIPI DSI-2 video data type value configured in IP.
0x0C	FRAME_WIDTH_PIXELS	RO	0x0080	15:0	Frame width (pixels per video line) configured in IP.
0x10	BYTES_PER_LANE	RO	0x02	4:0	PPI width in bytes configured in IP.
0x14	FEATURE_ENABLE	RO	0x03	1	ECC insertion configured in IP, 1 = enabled.
				0	CRC insertion configured in IP, 1 = enabled.
0x18	FRAME_COUNT	RO	0x00	31:0	Counter incremented on each video frame start sent.
<i>continued...</i>					

Address	Register Name	Access	Default	Bits	Description
0x1C	LINE_COUNT	RO	0x00	31:0	Counter incremented on each video line start sent.
0x20	PIXEL_COUNT	RO	0x00	31:0	Counter incremented on each video pixel sent.
0x24	COLLISION	RO	0x00	1:0	Internal collision detection. If either of these bits is set then an internal fault has occurred.

8.2. CVO and Video Timing Generator Registers

The registers within the Clocked Video Output (CVO) and Video Timing Generator are shown in the following tables. These are 32-bit word offsets from the base address of the MIPI DSI-2 instance; therefore, multiply these by four for byte offsets. Register writes should not change any bits marked as reserved from their previous value.

Note: Note the difference between the addresses for IP version 1.0.0 and 2.0.0. Make sure you use the correct address for the IP version in your design.

For more details on the functions controlled by the CVO registers, refer to the *About the Clocked Video Output IP* in the *Altera Video & Vision Processing Suite IP User Guide*.

For more details on the functions controlled by the Video Timing Generator registers, refer to *Video Timing Generator IP* in the *Altera Video & Vision Processing Suite IP User Guide*.

Related Information

- [About the Clocked Video Output IP](#)
- [Video Timing Generator IP](#)

8.2.1. Register Map for IP version 1.0.0

Table 30. CVO and Video Timing Generator Register Map for IP version 1.0.0

Address	Register Name	Access	Description
Clocked Video Output Parameterization Registers			
0x1000	VIP_PID	RO	Read this register to retrieve clocked video output IP product ID.
0x1010	VERSION	RO	Read this register to retrieve the version information for the Clocked Video Output IP.
0x1020	VID_FIFO_DEPTH	RO	The depth of the video input CDC FIFO.
0x1040	BPS	RO	The bits per video sample.
0x1070	PIXELS_IN_PARALLEL	RO	The number of pixels per clock transferred on the AXI4-S buses.
0x1080	NUMBER_OF_COLOR_PLANES	RO	The number of color planes per pixel.
0x10A0	CPU_CLK_FREQ_HZ	RO	The frequency of the processor interface clock in Hz.
<i>continued...</i>			

Address	Register Name	Access	Description
0x10BA	TIM_DIMENSIONS	RO	If set to 1, the processor interface diagnostics include the dimensions of the input timing reference. If set to 0, the dimensions of the timing reference are not available to the processor interface.
0x10C0	VID_IS_ASYNC	RO	If set to 1, the video input is treated as asynchronous to the timing reference. If set to 0, the video input must use the AXI4-S full-raster video clock.
0x10D0	VID_DIMENSIONS	RO	If set to 1, the processor interface diagnostics include the dimensions of the video input. If set to 0, the dimensions of the video input are not available to the processor interface.
0x10E0	VID_DEBUG	RO	If set to 1, the processor interface diagnostics include the locked, size err, and stall counters for the video input. If set to 0, the counters are not available to the processor Interface.
0x1130	HSIZE	RO	The number of bits used to build the horizontal counters and comparators inside the Video Timing Generator. This determines the maximum width of raster that can be generated.
0x1140	VSIZE	RO	The number of bits used to build the vertical counters and comparators inside the Video Timing Generator. This determines the maximum height of raster that can be generated.
0x1150	BUILD_HARD_FRAME_LOCK	RO	If set to true, the internal Video Timing Generator turns on the hard frame lock function. If set to false, the internal Video Timing Generator turns off the hard frame lock.
0x1160	BUILD_SOFT_FRAME_LOCK	RO	If set to true, the internal Video Timing Generator turns on the soft frame lock function. If set to false, the internal Video Timing Generator turns off the soft frame lock.
Clocked Video Output Core Registers			
0x1500	REG_STATUS	RO	Returns the status of the merge process.
0x1510	REG_BLACK_0	RW	The initial value of "black" for color plane 0.
0x1520	REG_BLACK_1	RW	The initial value of "black" for color plane 1.
0x1530	REG_BLACK_2	RW	The initial value of "black" for color plane 2.
0x1550	REG_FALLBACK	RW	Defines the behavior of the merge block if the video input fails.
0x15A0	REG_VID_DIMS	RO	The active height and width of the video input.
0x15B0	REG_VID_FREQ_DIMS	RO	The frequency of the video input clock in Hz.
0x15C0	REG_VID_CLKS_DIMS	RO	The number of video input clocks per input frame.
0x1600	REG_VID_LOCKED_COUNT	RO	Diagnostic counter that increments once at the start of each locked video frame.
0x1610	REG_VID_SIZE_ERR_COUNT	RO	Diagnostic counter that increments every time a mismatch between the video input and timing input is detected.
continued...			

Address	Register Name	Access	Description
0x1620 0x588	REG_VID_STALL_COUNT	RO	Diagnostic counter that increments every time the video input stalls and is dropped by the merge block.
Video Timing Generator Registers			
0x1900	REG_STATUS	RO	Reserved.
0x1910	REG_COMMIT	RW	Update internal parameters with new video standard.
0x1920	REG_MODE	RW	Control mode of operation.
0x1930	REG_RESET_POS	RW	Expected position of the start of the frame input signal relative to the raster.
0x1940	REG_TOTALS	RW	Total height and width of the raster.
0x1950	REG_HB_END	RW	First active pixel of a line.
0x1960	REG_V1B_POS	RW	Start and end of vertical blanking for field 1.
0x1970	REG_V2B_POS	RW	Start and end of vertical blanking for field 2.
0x1980	REG_FIELD_STARTS	RW	First lines of field 1 and 2.
0x1990	REG_HS_POS	RW	The start and end of horizontal sync.
0x19A0	REG_V1S_START	RW	The horizontal and vertical position of the start of the vertical sync for field 1.
0x19B0	REG_V1S_END	RW	The horizontal and vertical position of the end of the vertical sync for field 1.
0x19C0	REG_V2S_START	RW	The horizontal and vertical position of the start of the vertical sync for field 2.
0x19D0	REG_V2S_END	RW	The horizontal and vertical position of the end of the vertical sync for field 2.
0x19E0	REG_JITTER_CONT	RW	Timing parameters for the hard and soft frame lock.
0x19F0	REG_BLACK_0	RW	The initial value of "black" for color plane 0.
0x1A00	REG_BLACK_1	RW	The initial value of "black" for color plane 1.
0x1A10	REG_BLACK_2	RW	The initial value of "black" for color plane 2.
0x1A30	REG_FRAME_COUNTS	RO	Returns the total number of frames output and the number of external frame starts received.
0x1A40	REG_FRAME_LENGTH	RO	The number of video clocks between consecutive frame start input signals.
0x1A50	REG_VTOTAL_ADJ	RO	The total height of the raster after adjustment for soft lock.
0x1A60	REG_VID_FREQ	RO	The frequency of the video clock in Hz.
0x1A70	REG_GENLOCK_STATS0	RO	
0x1A80	REG_GENLOCK_STATS1	RO	Diagnostics for soft frame lock. Diagnostics for hard and soft frame lock.

8.2.2. Register Map for IP version 2.0.0

Table 31. CVO and Video Timing Generator Register Map for IP version 2.0.0

Address	Register Name	Access	Description
Clocked Video Output Parameterization Registers			
0x400	VIP_PID	RO	Read this register to retrieve clocked video output IP product ID.
0x404	VERSION	RO	Read this register to retrieve the version information for the Clocked Video Output IP.
0x408	VID_FIFO_DEPTH	RO	The depth of the video input CDC FIFO.
0x410	BPS	RO	The bits per video sample.
0x41C	PIXELS_IN_PARALLEL	RO	The number of pixels per clock transferred on the AXI4-S buses.
0x420	NUMBER_OF_COLOR_PLANES	RO	The number of color planes per pixel.
0x428	CPU_CLK_FREQ_HZ	RO	The frequency of the processor interface clock in Hz.
0x42C	TIM_DIMENSIONS	RO	If set to 1, the processor interface diagnostics include the dimensions of the input timing reference. If set to 0, the dimensions of the timing reference are not available to the processor interface.
0x430	VID_IS_ASYNC	RO	If set to 1, the video input is treated as asynchronous to the timing reference. If set to 0, the video input must use the AXI4-S full-raster video clock.
0x434	VID_DIMENSIONS	RO	If set to 1, the processor interface diagnostics include the dimensions of the video input. If set to 0, the dimensions of the video input are not available to the processor interface.
0x438	VID_DEBUG	RO	If set to 1, the processor interface diagnostics include the locked, size err, and stall counters for the video input. If set to 0, the counters are not available to the processor Interface.
0x44C	HSIZE	RO	The number of bits used to build the horizontal counters and comparators inside the Video Timing Generator. This determines the maximum width of raster that can be generated.
0x450	VSIZE	RO	The number of bits used to build the vertical counters and comparators inside the Video Timing Generator. This determines the maximum height of raster that can be generated.
0x454	BUILD_HARD_FRAME_LOCK	RO	If set to true, the internal Video Timing Generator turns on the hard frame lock function. If set to false, the internal Video Timing Generator turns off the hard frame lock.
0x458	BUILD_SOFT_FRAME_LOCK	RO	If set to true, the internal Video Timing Generator turns on the soft frame lock function. If set to false, the internal Video Timing Generator turns off the soft frame lock.
Clocked Video Output Core Registers			
0x540	REG_STATUS	RO	Returns the status of the merge process.
continued...			

Address	Register Name	Access	Description
0x544	REG_BLACK_0	RW	The initial value of "black" for color plane 0.
0x548	REG_BLACK_1	RW	The initial value of "black" for color plane 1.
0x54C	REG_BLACK_2	RW	The initial value of "black" for color plane 2.
0x554	REG_FALLBACK	RW	Defines the behavior of the merge block if the video input fails.
0x568	REG_VID_DIMS	RO	The active height and width of the video input.
0x56C	REG_VID_FREQ_DIMS	RO	The frequency of the video input clock in Hz.
0x570	REG_VID_CLKS_DIMS	RO	The number of video input clocks per input frame.
0x580	REG_VID_LOCKED_COUNT	RO	Diagnostic counter that increments once at the start of each locked video frame.
0x584	REG_VID_SIZE_ERR_COUNT	RO	Diagnostic counter that increments every time a mismatch between the video input and timing input is detected.
0x588	REG_VID_STALL_COUNT	RO	Diagnostic counter that increments every time the video input stalls and is dropped by the merge block.
Video Timing Generator Registers			
0x640	REG_STATUS	RO	Reserved.
0x644	REG_COMMIT	RW	Update internal parameters with new video standard.
0x648	REG_MODE	RW	Control mode of operation.
0x64c	REG_RESET_POS	RW	Expected position of the start of the frame input signal relative to the raster.
0x650	REG_TOTALS	RW	Total height and width of the raster.
0x654	REG_HB_END	RW	First active pixel of a line.
0x658	REG_V1B_POS	RW	Start and end of vertical blanking for field 1.
0x65c	REG_V2B_POS	RW	Start and end of vertical blanking for field 2.
0x660	REG_FIELD_STARTS	RW	First lines of field 1 and 2.
0x664	REG_HS_POS	RW	The start and end of horizontal sync.
0x668	REG_V1S_START	RW	The horizontal and vertical position of the start of the vertical sync for field 1.
0x66c	REG_V1S_END	RW	The horizontal and vertical position of the end of the vertical sync for field 1.
0x670	REG_V2S_START	RW	The horizontal and vertical position of the start of the vertical sync for field 2.
0x674	REG_V2S_END	RW	The horizontal and vertical position of the end of the vertical sync for field 2.
0x678	REG_JITTER_CONT	RW	Timing parameters for the hard and soft frame lock.
0x67c	REG_BLACK_0	RW	The initial value of "black" for color plane 0.
<i>continued...</i>			

Address	Register Name	Access	Description
0x680	REG_BLACK_1	RW	The initial value of "black" for color plane 1.
0x684	REG_BLACK_2	RW	The initial value of "black" for color plane 2.
0x68c	REG_FRAME_COUNTS	RO	Returns the total number of frames output and the number of external frame starts received.
0x690	REG_FRAME_LENGTH	RO	The number of video clocks between consecutive frame start input signals.
0x694	REG_VTOTAL_ADJ	RO	The total height of the raster after adjustment for soft lock.
0x698	REG_VID_FREQ	RO	The frequency of the video clock in Hz.
0x69c	REG_GENLOCK_STATS0	RO	Diagnostics for hard and soft frame lock.
0x6a0	REG_GENLOCK_STATS1	RO	Diagnostics for soft frame lock.



9. Document Revision History for the MIPI DSI-2 IP User Guide

Document Version	Quartus Prime Version	IP Version	Changes
2025.11.21	25.3	2.0.0	Initial release.